t3-w02 — Plane Shapes (2-D Shapes) — Plane shapes: types, properties and perimeter

Run `c0c3-deepseek-native` · conditions C0, C1, C2, C3 · the pack shown is draw 3 of each.

HER LESSON PLAN

Plane Shapes (2-D Shapes) — Plane shapes: types, properties and perimeter

Objectives

1. identify the types of plane shapes and state their properties, similarities and differences
2. identify the parts of a circle
3. find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Step I — intro: angles (previous knowledge)

Teacher displays the drawings of angles on the board asking from the students at random the various angles while the students respond to teachers questions. Teacher adopts the students answers if correct, otherwise teacher corrects the student and the lesson progresses.

- **teacher** displays the drawings of angles on the board, asking the students at random the various angles
- **student** respond to the teacher's questions
- **teacher** adopts the students' answers if correct, otherwise corrects the student

Step II — types of plane shapes, their properties, similarities and differences

Teacher draws the various shapes on the board while students identify their properties. The teacher writes the properties of the shapes on the board for the students to learn.

- **teacher** draws the various shapes on the board
- **student** identify their properties
- **teacher** writes the properties of the shapes on the board for the students to learn

Step III — parts of a circle

Teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings.

- **teacher** explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings

Step IV — perimeter of plane shapes

Teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given.

- **teacher** works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given

Evaluation

Teacher test the students understanding on the topic by giving them some question to solve

1. Find the perimeter of a square of sides 8cm
2. Find the perimeter of a rectangle of length 7cm and breadth 4cm.
3. What is the perimeter of regular pentagon with sides 5cm.
4. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?
5. Find the perimeter of a parallelogram whose base is 6cm and sides 5cm
6. The circumference of a circle is 22cm – Find the radius of a circle $(\pi = \frac{22}{7})$.

7. A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle.
8. Calculate the perimeter of the figure below. \begin{center} \begin{tikzpicture}[scale=0.42] \draw (0,0)--(9,0)--(9,5)--(8,5) arc[start angle=0, end angle=-180, radius=3.5]--(0,5)--cycle; \node[above] at (0.5,5) {1cm}; \node[above] at (8.5,5) {1cm}; \node[right] at (9,2.5) {5cm}; \node[below] at (4.5,0) {9cm}; \end{tikzpicture} \end{center}
9. Find the perimeter of a regular hexagon with sides 5cm.

HER LESSON NOTE

Verbatim, as transcribed. **C2 and C3 receive this; C0 and C1 never do** — which is the whole of what separates them.

Step II — types of plane shapes, their properties, similarities and differences

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#### Hints

Plane shapes are also known as two dimensional shapes (2-D shapes). They include: triangle,
rectangle, squares, parallelogram, trapezium, circle and other polygons.

#### Triangles

Isosceles: a single tick on each of the two slanted sides, an arc at each base angle. Scalene:
one, two and three ticks on the three sides, an arc at each vertex. Equilateral: a single tick
on each of the three sides, an arc at each vertex.

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(4,0)--(2,4)--cycle;
  \draw (0.8,1.6)--(1.2,1.4);
  \draw (3.2,1.6)--(2.8,1.4);
  \draw (0,0)++(0:0.7) arc[start angle=0, end angle=63, radius=0.7];
  \draw (4,0)++(180:0.7) arc[start angle=180, end angle=117, radius=0.7];
  \node[below] at (2,-0.6) {Isosceles};
  \begin{scope}[xshift=7cm]
    \draw (0,0)--(5,0)--(1.4,3.4)--cycle;
    \draw (0.38,1.83)--(1.02,1.57);
    \draw (3.02,1.87)--(3.50,2.38);
    \draw (3.38,1.53)--(3.86,2.04);
    \foreach \d in {-0.25,0,0.25} { \draw (2.5+\d,-0.25)--(2.5+\d,0.25); }
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=68, radius=0.7];
    \draw (5,0)++(180:0.8) arc[start angle=180, end angle=137, radius=0.8];
    \draw (1.4,3.4)++(-113:0.7) arc[start angle=-113, end angle=-25, radius=0.7];
    \node[below] at (2.5,-0.6) {Scalene};
  \end{scope}
  \begin{scope}[xshift=14cm]
    \draw (0,0)--(4,0)--(2,3.46)--cycle;
    \draw (0.8,1.55)--(1.2,1.32);
    \draw (3.2,1.55)--(2.8,1.32);
    \draw (1.8,-0.25)--(2.2,0.25);
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=60, radius=0.7];
    \draw (4,0)++(180:0.7) arc[start angle=180, end angle=120, radius=0.7];
    \draw (2,3.46)++(-120:0.7) arc[start angle=-120, end angle=-60, radius=0.7];
    \node[below] at (2,-0.6) {Equilateral};
  \end{scope}
\end{tikzpicture}
\end{center}

| Isosceles | Scalene | Equilateral |
|---|---|---|
| (i) 2 adjacent side are equal | No side equal | 3 sides are equal |
| (ii) base angles are equal | No angle is equal to the other | All angles are equal ($60^\circ$
each) |
| (iii) one line of symmetry | No line of symmetry. | 3 lines of symmetry. |

#### Squares and Rectangles
```

Square: a right-angle mark at each corner, a single tick on each of the four sides. Rectangle: a right-angle mark at each corner, a single tick on each of the two longer sides and a double tick on each of the two shorter sides.

```
\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (3.4,3.4);
  \draw (0,0.5)--(0.5,0.5)--(0.5,0);
  \draw (3.4,0.5)--(2.9,0.5)--(2.9,0);
  \draw (0,2.9)--(0.5,2.9)--(0.5,3.4);
  \draw (3.4,2.9)--(2.9,2.9)--(2.9,3.4);
  \draw (1.7,-0.25)--(1.7,0.25);
  \draw (1.7,3.15)--(1.7,3.65);
  \draw (-0.25,1.7)--(0.25,1.7);
  \draw (3.15,1.7)--(3.65,1.7);
  \node[below] at (1.7,-0.9) {Squares};
  \begin{scope}[xshift=8cm]
    \draw (0,0) rectangle (5,3);
    \draw (0,0.5)--(0.5,0.5)--(0.5,0);
    \draw (5,0.5)--(4.5,0.5)--(4.5,0);
    \draw (0,2.5)--(0.5,2.5)--(0.5,3);
    \draw (5,2.5)--(4.5,2.5)--(4.5,3);
    \draw (2.5,-0.25)--(2.5,0.25);
    \draw (2.5,2.75)--(2.5,3.25);
    \draw (-0.25,1.35)--(0.25,1.35); \draw (-0.25,1.65)--(0.25,1.65);
    \draw (4.75,1.35)--(5.25,1.35); \draw (4.75,1.65)--(5.25,1.65);
    \node[below] at (2.5,-0.9) {Rectangles};
  \end{scope}
\end{tikzpicture}
\end{center}
```

Squares	Rectangles
i) All sides are equal	2 opposite sides are equal
(ii) 2 pairs of parallel sides	2 pair of parallel sides.
(iii) All angles are equal (90°) each.	All angles are equal (90°) each.
(iv) Four lines of symmetry	Two lines of symmetry
(v) Diagonals bisect it at right angle.	Diagonals bisect it but not at right angle.

Parallelograms and Trapezium

Parallelogram: a double arrow on each of the two horizontal sides, a single arrow on each of the two slanted sides. Trapezium: a single arrow on each of the two parallel sides, and one diagonal drawn in.

```
\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(5,0)--(6.6,3)--(1.6,3)--cycle;
  \draw[->] (2.4,0)--(2.6,0);
  \draw[->] (3.9,3)--(4.1,3);
  \draw[->] (5.7,1.4)--(5.85,1.65);
  \draw[->] (5.85,1.4)--(6.0,1.65);
  \draw[->] (0.7,1.4)--(0.85,1.65);
  \draw[->] (0.85,1.4)--(1.0,1.65);
  \node[below] at (3,-0.6) {Parallelogram};
  \begin{scope}[xshift=10cm]
    \draw (0,0)--(6,0)--(4.6,3)--(1.6,3)--cycle;
    \draw (0,0)--(4.6,3);
    \draw[->] (2.9,0)--(3.1,0);
    \draw[->] (3.0,3)--(3.2,3);
```

```
    \node[below] at (3,-0.6) {Trapezium};
  \end{scope}
\end{tikzpicture}
\end{center}

| Parallelogram | Trapezium |
|---|---|
| i) 2 pairs of opposite side are equal and parallel | Only a pair of sides are parallel |
| (ii) opposite angles are equal | opposite angles are not equal |
| (iii) No line of symmetry | No line of symmetry |
| (iv) Diagonals divide it into equal triangles | Diagonal divide it into two unequal triangles.
|

#### Polygons

A polygon is a closed shape that has three or more straight sides.

| Name of Polygon | No of sides | No of triangle ($n-2$) |
|---|---|---|
| Triangle | 3 | 1 |
| Quadrilateral | 4 | 2 |
| Pentagon | 5 | 3 |
| Hexagon | 6 | 4 |
| Heptagon | 7 | 5 |
| Octagon | 8 | 6 |
| Nonagon | 9 | 7 |
| Decagon | 10 | 8 |
| Hendecagon or undecagon | 11 | 9 |
| Dodecagon | 12 | 10 |
```

Step III — parts of a circle

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#### Parts of a circle

Two labelled circles. The first has the diameter drawn across it and a radius from its centre to
the circumference, with the arc and the circumference labelled. The second has a chord near the
top with the segment above it shaded, a shaded sector below with its right-hand bounding line
labelled Radius, and a tangent touching it on the right.

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (-3,0)--(3,0);
  \node[above] at (0,0.1) {Diameter};
  \draw (0,0)--(0,-3);
  \node[right] at (0.45,-1.9) {Radius};
  \draw[->] (-5.9,-3.4)--(-2.3,-1.9);
  \node[left] at (-5.9,-3.4) {arc};
  \draw[->] (4.4,3.4)--(2.2,2.2);
  \node[right] at (4.4,3.4) {Circumference};
  \begin{scope}[xshift=17cm]
    \draw (0,0) circle (3);
    \draw (-2.29,1.94)--(2.29,1.94);
    \begin{scope}
      \clip (0,0) circle (3);
      \clip (-3,1.94) rectangle (3,3);
      \foreach \x in {-3,-2.7,...,3} { \draw (\x,1.8)--(\x+1.2,3.0); }
    \end{scope}
    \node[below] at (0,1.9) {Chord};
```

```
    \draw[->] (3.2,4.2)--(0.6,2.5);
    \node[right] at (3.2,4.2) {Segment};
    \begin{scope}
      \clip (0,0)--(-80:3) arc[start angle=-80, end angle=-25, radius=3]--cycle;
      \foreach \y in {-3.4,-3.1,...,0} { \draw (0,\y)--(3,\y+1.2); }
    \end{scope}
    \draw (0,0)--(-80:3);
    \draw (0,0)--(-25:3) node[midway, sloped, above] {Radius};
    \draw[->] (-3.6,-4.2)--(1.0,-2.0);
    \node[left] at (-3.6,-4.2) {Sector};
    \draw (3,-2.4)--(3,2.4);
    \node[right] at (3.2,0) {Tangent};
  \end{scope}
\end{tikzpicture}
\end{center}

- **Arc:** This is a curved part of the circumference.
- **Radius:** This is any straight line joining the centre of the circle to the circumference.
- **Chord:** This is any straight line joining two point on the circumference of a circle.
- **Diameter:** Any chord that passes through the centre of the circle.
```

Step IV — perimeter of plane shapes

```
#### Perimeter

Perimeter is the distance round a shape.

**Worked examples**

##### 1. Calculate the perimeter of a square of side 15cm.

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (5,5);
  \node[left]  at (-1.4,4.4) {Square};
  \node[above] at (2.5,5) {$L$\quad 15cm};
  \node[below] at (2.5,0) {$L$\quad 15cm};
  \node[left]  at (0,3) {15cm};
  \node[left]  at (0,1.8) {$L$};
  \node[right] at (5,3) {15cm};
  \node[right] at (5,1.8) {$L$};
\end{tikzpicture}
\end{center}

**Solution**

Perimeter of a square $= L + L + L + L = 4L$

$= 15 + 15 + 15 + 15 = 4(15)$

$= 60\mathrm{cm}$

##### 2. Find the perimeter of a rectangle whose length 10cm and breadth 7cm.

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4.5);
  \node[above] at (3.5,4.5) {$L$\quad 10cm};
  \node[below] at (3.5,0) {$L$};
```

```
  \node[left]  at (0,2.25) {$b$};
  \node[right] at (7,2.7) {$b$};
  \node[right] at (7,1.6) {7cm};
\end{tikzpicture}
\end{center}

**Solution**

Perimeter of rectangle $= L + b + L + b$

$= L + L + b + b$

$= 2L + 2b$

$= 2(L + B)$

$= 10\mathrm{cm} + 7\mathrm{cm} + 10\mathrm{cm} + 7\mathrm{cm}$

$= 10\mathrm{cm} + 10\mathrm{cm} + 7\mathrm{cm} + 7\mathrm{cm}$

$= 2(10) + 2(7)$

$= 2(10 + 7)$

$= 2(17)$

$= 34\mathrm{cm}$

##### 3. If the perimeter of a rectangle is 60cm and its length is 17cm. Calculate its breadth.

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4);
  \node[above] at (3.5,4) {$L = 17$cm};
  \node[below] at (3.5,0) {$L = 17$cm};
  \node[left]  at (0,2) {$b$};
  \node[right] at (7,2) {$b$};
\end{tikzpicture}
\end{center}

**Solution**

Perimeter $= 60\mathrm{cm}$

$P = L + b + L + b$

$60 = 17\mathrm{cm} + b + 17\mathrm{cm} + b$

$60 = 17\mathrm{cm} + 17\mathrm{cm} + b + b$

$60\mathrm{cm} = 2(17\mathrm{cm}) + 2b$

$60\mathrm{cm} = 2(17\mathrm{cm} + b)$

$60\mathrm{cm} = 34\mathrm{cm} + 2b$

collect like terms.

$60\mathrm{cm} - 34\mathrm{cm} = 2b$
```

```
$2b = 26\mathrm{cm}$

Divide both side by 2

$\frac{2b}{2} = \frac{26\mathrm{cm}}{2}$

$b = 13\mathrm{cm}$

##### 4. Find the perimeter of the trapezium below.

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(7.5,0)--(6.0,3.2)--(2.0,3.2)--cycle;
  \node[below] at (3.75,0) {$a$\quad 11cm};
  \node[above] at (4.0,3.2) {$c$\quad 5cm};
  \node[left]  at (0.95,2.3) {9cm};
  \node[left]  at (0.65,1.35) {$b$};
  \node[right] at (6.9,2.3) {4cm};
  \node[right] at (7.15,1.35) {$d$};
\end{tikzpicture}
\end{center}

Perimeter of trapezium $= a + b + c + d$

$= 11\mathrm{cm} + 9\mathrm{cm} + 5\mathrm{cm} + 4\mathrm{cm}$

$= 29\mathrm{cm}$

##### 5. Find the perimeter of the parallelogram below.

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(6,0)--(8,3.2)--(2,3.2)--cycle;
  \draw[->] (2.7,0)--(2.9,0);
  \draw[->] (4.7,3.2)--(4.9,3.2);
  \draw[->] (0.9,1.5)--(1.05,1.75);
  \draw[->] (6.9,1.5)--(7.05,1.75);
  \node[below] at (3,0) {9cm\quad $a$};
  \node[above] at (5,3.2) {$a$};
  \node[left]  at (0.9,1.6) {$b$};
  \node[right] at (7.2,1.9) {$b$};
  \node[right] at (7.4,1.1) {5cm};
\end{tikzpicture}
\end{center}

**Solution**

Perimeter of parallelogram $= a + a + b + b$

$= 2a + 2b$

$= 2(a + b)$

$= 2(9 + 5)$

$= 2(14)$

$= 28\mathrm{cm}$

##### 6. Find the perimeter of a regular hexagon sides 5.2cm.
```

```
**Solution.**

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \foreach \i in {0,...,5} { \coordinate (H\i) at ({\i*60}:3); }
  \draw (H0)--(H1)--(H2)--(H3)--(H4)--(H5)--cycle;
  \node[above]       at (0,4.0) {Regular hexagon};
  \node[above]       at (0,2.65) {5.2cm};
  \node[above right] at (2.4,1.3) {5.2cm};
  \node[below right] at (2.4,-1.3) {5.2cm};
  \node[below]       at (0,-2.65) {5.2cm};
  \node[below left]  at (-2.4,-1.3) {5.2cm};
  \node[above left]  at (-2.4,1.3) {5.2cm};
\end{tikzpicture}
\end{center}

Hexagon has 6 sides.

Perimeter $= 6 \times 5.2\mathrm{cm}$

$= 31.2\mathrm{cm}$

##### 7. Calculate the perimeter of circle of radius 7cm (use the value $3\frac{1}{7}$ for $\pi$)

**Solution**

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (0,0)--(3,0) node[midway,above] {7cm};
\end{tikzpicture}
\end{center}

Perimeter of a circle $= 2\pi r$ or $\pi d$

$= 2 \times 3\frac{1}{7} \times 7\mathrm{cm}$

$= 2 \times \frac{22}{7} \times 7$

$= 2 \times 22 = 44\mathrm{cm}$

using $\pi d$.

Since $r = 7\mathrm{cm}$

$d = 2 \times r = 2 \times 7\mathrm{cm} = 14\mathrm{cm}$

$\frac{22}{7} \times 14\mathrm{cm}$

$= 22 \times 2$

$= 44\mathrm{cm}$

##### 8. A rope is wound 50 times round a cylinder of radius 25cm. How long is the rope? $\pi = 3.14$.

**Solution**

\begin{center}
```

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) ellipse (2 and 0.6);
  \draw (-2,0)--(-2,-6);
  \draw (2,0)--(2,-6);
  \draw (-2,-6) arc[start angle=180, end angle=360, x radius=2, y radius=0.6];
  \foreach \y in {-0.9,-1.8,-2.7,-3.6,-4.5,-5.4} {
    \draw (-2,\y) arc[start angle=180, end angle=360, x radius=2, y radius=0.55];
    \draw (2,\y-0.03)--(-2,\y-0.9);
  }
\end{tikzpicture}
\end{center}

length of the rope $=$ circumference.

Circumference $= 2\pi r$

$= 2 \times 3.14 \times 25$

$= 157\mathrm{cm}$

Total length of the rope $= 157\mathrm{cm} \times 50$

$= 7850\mathrm{cm}$.

##### 9. The circumference of a circle is 440cm. Find the radius $(\pi = \frac{22}{7})$

**Solution**

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (0,0)--(3,0) node[midway,above] {$r$};
\end{tikzpicture}
\end{center}

Circumference $= 2\pi r$

$440 = 2 \times \frac{22}{7} \times r$

$440 = \frac{44r}{7}$

cross multiply.

$440 \times 7 = 44r$

$3080 = 44r$

divide both side by $44$

$\frac{44r}{44} = \frac{3080}{44}$

$r = 70\mathrm{cm}$

##### 10. Calculate the perimeter of the shape below.

\begin{center}
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(0,5)--(0,5) arc[start angle=90, end angle=0, radius=5]--(0,0);
  \node[left]  at (0,2.5) {4.9 cm};
  \node[below] at (2.5,0) {4.9cm};
```

```
\end{tikzpicture}
\end{center}

**Solution**

length of the curved part $= \frac{1}{4} \times 2\pi r$

$= \frac{1}{4} \times 2 \times \frac{22}{7} \times 4.9$

$= \frac{1}{4} \times \frac{44}{7} \times 4.9$

$= \frac{11}{7} \times \frac{49}{10}$

$= \frac{11 \times 7}{10}$

$= \frac{77}{10}$

$= 7.7\mathrm{cm}$

Perimeter of the shape $= 4.9 + 4.9 + 7.7$

$= 17.5\mathrm{cm}$
```

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Plane Shapes (2-D Shapes)

`subtopic`

Plane shapes: types, properties and perimeter

`term`

Third Term

`week`

2

`lesson_objectives`

• identify the types of plane shapes and state their properties, similarities and differences
• identify the parts of a circle
• find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

`objective_text`

identify the types of plane shapes and state their properties, similarities and differences

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — identify the parts of a circle

`objective_text`

identify the parts of a circle

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`objective_text`

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

All About Plane Shapes

EXPLANATION

A **plane shape** (also called a 2-D shape) is a flat shape that lies on a flat surface, like a sheet of paper. Every plane shape has only **length and width** — it has no thickness. Around you, the face of a ruler, the top of your desk, and the face of a wall clock are all plane shapes.

Plane shapes come in two big families.

1. Regular shapes. A shape is **regular** when every side is the same length and every angle is the same size. Examples are the **square** (4 equal sides, 4 right angles) and the **equilateral triangle** (3 equal sides, 3 equal angles).

2. Irregular shapes. A shape is **irregular** when its sides are not all the same length or its angles are not all the same size. Examples are the **rectangle** (opposite sides equal, all angles right angles) and the **scalene triangle** (no two sides equal).

Let's look at the common plane shapes and their properties side by side.

Shape	Number of sides	Properties
Triangle	3	3 sides and 3 angles; the three angles add up to 180°
Square	4	All 4 sides equal; all 4 angles are right angles (90°); both pairs of opposite sides are parallel
Rectangle	4	Opposite sides equal; all 4 angles are right angles; both pairs of opposite sides are parallel
Trapezium	4	One pair of opposite sides parallel; the other pair is not parallel
Parallelogram	4	Both pairs of opposite sides parallel and equal; opposite angles equal
Circle	0 (curved)	One curved line; every point on the curve is the same distance from the centre

Similarities and differences. The square, rectangle, trapezium and parallelogram all have **four sides**, so they are all **quadrilaterals**. They differ in which sides are equal and which are parallel: a square has all sides equal, a rectangle has only opposite sides equal, a parallelogram has opposite sides equal and both pairs parallel, and a trapezium has only one pair of parallel sides.

A square and a rectangle are alike because both have four right angles and both pairs of opposite sides parallel. They differ because the square's four sides are all equal, while a rectangle's opposite sides are equal but its four sides are not all equal.

The triangle stands apart because it has three sides, and the circle stands apart because it has no straight sides at all — it is made of one continuous curved line. So, to sort a shape, ask two questions: *How many sides does it have?* and *Are the sides equal and are any sides parallel?*

WORKED_EXAMPLE

Example 1: Name the shape and give one property.
A shape has four sides, all four sides are equal, and all four angles are right angles. What is it, and what family does it belong to?

Solution. Four sides with all sides equal and all angles right angles is a **square**. Because it has four sides, it is a **quadrilateral**. Its property is that both pairs of opposite sides are parallel.

Example 2: Compare two shapes.
State one way a parallelogram and a trapezium are similar and one way they differ.

Solution. **Similar:** both have four sides, so both are quadrilaterals.
Differ: a parallelogram has *both* pairs of opposite sides parallel, while a trapezium has only *one* pair of parallel sides.

Example 3: Tell one shape from another.
A shape has four sides, only opposite sides are equal, and all angles are right angles. Is it a square or a rectangle? Explain.

Solution. It is a **rectangle**. A square needs *all four* sides equal, but this shape has only *opposite* sides equal. A rectangle has opposite sides equal and four right angles, which matches exactly.

1. Write down the number of sides of (a) a triangle, (b) a square, (c) a circle.
answer: (a) 3 sides, (b) 4 sides, (c) 0 straight sides — a circle is one curved line.

2. A shape has four sides, both pairs of opposite sides are parallel, and all four angles are right angles. What shape is it? Give the special property that makes it different from a rectangle.

answer: It is a square. Its special property is that all four sides are equal (a rectangle only has opposite sides equal).

3. State one similarity and one difference between a parallelogram and a rectangle.

answer: Similar: both have four sides (both are quadrilaterals) and both have both pairs of opposite sides parallel. Difference: a rectangle has all angles equal (right angles), while a parallelogram need not have right angles.

4. A shape has one pair of opposite sides parallel and the other pair not parallel. Name the shape and say how it is different from a parallelogram.

answer: It is a trapezium. A parallelogram has both pairs of opposite sides parallel, but a trapezium has only one pair parallel.

5. A triangle has three equal sides. (a) What is this triangle called? (b) Is this triangle a regular or an irregular shape? Give a reason.

answer: (a) Equilateral triangle. (b) It is regular, because all three sides are equal and all three angles are equal.

Getting to Know the Circle and Its Parts

EXPLANATION

A circle is a plane shape drawn so that every point on its outline is the same distance from one fixed point in the middle. That fixed point is called the centre. Because the circle is one smooth, unbroken curve with no corners, it is different from shapes like the square, rectangle or triangle.

To talk about a circle clearly, we give names to its parts. Here are the ones you will use most often:

- The centre is the exact middle point of the circle. All the other parts are measured from it.
- The radius is any straight line from the centre to a point on the circle. Every radius in a circle is the same length. You can think of the radius as "the spoke of a wheel".
- The diameter is a straight line that passes through the centre and touches the circle at two points on opposite sides. The diameter is exactly twice the radius, so if the radius is 4 cm, the diameter is 8 cm.
- The circumference is the distance all the way around the circle. It is the circle's "perimeter" — the length of its curved outline.
- The chord is any straight line joining two points on the circle. Notice that the diameter is actually a special chord: it is the longest chord, because it passes through the centre.
- An arc is a part of the circle's curved outline, like a slice of the circumference.
- A sector is the part of the circle shaped like a slice of cake, bounded by two radii and the arc between them.
- A segment is the part of the circle cut off by a chord, like the sliver between the chord and the arc.
- A semicircle is exactly half of the circle, formed when a diameter cuts the circle into two equal halves.

A quick way to keep these straight: the radius and diameter are distances measured in the middle of the circle (lines), while the circumference, arc and semicircle describe the outline. The chord and the diameter both join two points on the circle — the difference is that the diameter must pass through the centre, and the chord need not.

WORKED_EXAMPLE

Let us look at a circle with its parts labelled, so you can see how the names fit together.

Imagine a circle whose centre is the point O. A point on the outline is marked A, and the point directly opposite it on the other side is marked B.

- The line from O to A is a radius. The line from O to B is another radius, and it is the same length as OA.
- The line from A to B passes through O, so it is a diameter. If OA is 5 cm, then AB (the diameter) is 5 cm × 2 = 10 cm.
- The whole curved outline of the circle is its circumference.
- If we draw a line from A to a third point C on the circle, that line AC is a chord (as long as it does not pass through the centre).
- The curved part of the outline between A and C is an arc.
- The slice bounded by OA, OC and the arc between A and C is a sector.
- The sliver between the chord AC and the arc AC is a segment.

So when you are asked to identify a part, ask yourself one question: is it a straight line through the centre (diameter), a straight line from the centre (radius), a straight line joining two points on the circle without passing through the centre (chord), or part of the curved outline (arc)? That one question sorts out almost every case.

1. A circle has a centre at point O, and a point P lies on the outline of the circle. A straight line is drawn from O to P. What is this line called, and what is its length called?

answer: It is a radius. Its length is the radius of the circle.

2. The radius of a circle is 6 cm. What is the length of its diameter? Explain your answer.

answer: 12 cm. The diameter is twice the radius, so 6 cm × 2 = 12 cm.

3. A straight line is drawn joining two points on a circle, but it does not pass through the centre. What is this line called?

answer: A chord.

4. The line AB passes through the centre of a circle and touches the circle at A and B. The line CD joins two points on the circle but does not pass through the centre. Name each of the lines AB and CD, and say which is longer.

answer: AB is a diameter and CD is a chord. AB is longer, because the diameter is the longest chord of a circle.

5. Match each part of a circle to what it is: (a) the distance all the way around the circle; (b) exactly half of the circle; (c) a part of the curved outline between two points; (d) the slice bounded by two radii and the arc between them.

answer: (a) circumference; (b) semicircle; (c) arc; (d) sector.

Objective 3

Finding the Perimeter of Plane Shapes

EXPLANATION

The **perimeter** of a shape is the total distance all the way around its outside edge. It is measured in units of length, such as centimetres (cm), metres (m) or kilometres (km). Think of walking once around the shape along its boundary — the distance you cover is its perimeter.

To find the perimeter of any plane (flat, 2-D) shape, you add up the lengths of all its sides. Here is how that works for the shapes in this lesson.

Square — all four sides are equal, so perimeter = 4 × side. If a square has sides of 5 cm, its perimeter is 4 × 5 = 20 cm.

Rectangle — opposite sides are equal, so there are two lengths and two widths. Perimeter = 2 × length + 2 × width, or (length + width) × 2. For a rectangle 8 cm long and 3 cm wide, perimeter = (8 + 3) × 2 = 11 × 2 = 22 cm.

Regular polygon — a regular polygon has all sides equal (a square is one example). So for any regular polygon, perimeter = number of sides × length of one side. A regular hexagon (6 sides) with each side 4 cm has perimeter 6 × 4 = 24 cm.

Trapezium — it has four sides, and no special rule beyond adding them all. Just add all four side lengths together.

Parallelogram — opposite sides are equal, just like a rectangle, so it has two "long" sides and two "short" sides. Perimeter = 2 × (long side + short side). A parallelogram with sides 9 cm and 5 cm has perimeter (9 + 5) × 2 = 28 cm.

Circle — the distance around a circle is its **circumference**. The formula uses **π** (pi), about 3.14. Circumference = π × diameter, or 2 × π × radius. The **diameter** is the distance straight across the circle through its centre, and the **radius** is half of that (from the centre to the edge). A circle with a diameter of 10 cm has circumference 3.14 × 10 = 31.4 cm.

Whichever shape you meet, the same idea applies: the perimeter is the total of the distances around the outside edge.

WORKED_EXAMPLE

Worked example 1 — a rectangle
A rectangular field is 40 m long and 25 m wide. Find its perimeter.
- A rectangle has two lengths and two widths.
- Perimeter = (length + width) × 2 = (40 + 25) × 2 = 65 × 2 = 130 m.
- So the perimeter of the field is **130 m**.

Worked example 2 — a regular polygon
A regular pentagon (a 5-sided shape with all sides equal) has each side 7 cm. Find its perimeter.
- Perimeter = number of sides × length of one side.
- Perimeter = 5 × 7 = 35 cm.
- So the perimeter is **35 cm**.

Worked example 3 — a trapezium
A trapezium has sides of 6 cm, 4 cm, 6 cm and 3 cm. Find its perimeter.
- A trapezium has no special shortcut, so just add the four sides.
- Perimeter = 6 + 4 + 6 + 3 = 19 cm.
- So the perimeter is **19 cm**.

Worked example 4 — a parallelogram
A parallelogram has a long side of 12 cm and a short side of 8 cm. Find its perimeter.
- Opposite sides are equal, so there are two 12 cm sides and two 8 cm sides.
- Perimeter = (long + short) × 2 = (12 + 8) × 2 = 20 × 2 = 40 cm.
- So the perimeter is **40 cm**.

Worked example 5 — a circle
A circular pond has a radius of 7 m. Find its circumference. Use π = 3.14.
- Circumference = 2 × π × radius.

> - Circumference = 2 × 3.14 × 7 = 6.28 × 7 = 43.96 m.
> - So the circumference is **43.96 m**.

1. A square garden has sides of 9 m. What is its perimeter?
answer: 36 m

2. A rectangular board is 15 cm long and 6 cm wide. Find its perimeter.
answer: 42 cm

3. A regular octagon (8 sides) has each side 5 cm. What is its perimeter?
answer: 40 cm

4. A trapezium has sides of length 10 cm, 7 cm, 10 cm and 4 cm. Find its perimeter.
answer: 31 cm

5. A parallelogram has a long side of 14 cm and a short side of 9 cm. Find its perimeter.
answer: 46 cm

6. A circular wheel has a diameter of 20 cm. Find its circumference. Use π = 3.14.
answer: 62.8 cm

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Plane Shapes (2-D Shapes)

`subtopic`

Plane shapes: types, properties and perimeter

`term`

Third Term

`week`

2

`lesson_objectives`

• identify the types of plane shapes and state their properties, similarities and differences
• identify the parts of a circle
• find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`entry_behaviour`

• Students have been taught angles.

`instructional_materials`

• Cardboard containing angles presentation.

Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

`objective_text`

identify the types of plane shapes and state their properties, similarities and differences

`current_step`

1

`instructional_events`

- present_content
- elicit_performance
- present_content

`assessment_points`

- How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — identify the parts of a circle

`objective_text`

identify the parts of a circle

`current_step`

2

`instructional_events`

- present_content

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`objective_text`

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`current_step`

3

`assessment_points`

- Find the perimeter of a square of sides 8cm
- Find the perimeter of a rectangle of length 7cm and breadth 4cm.
- What is the perimeter of regular pentagon with sides 5cm.
- Find the perimeter of a parallelogram whose base is 6cm and sides 5cm
- The circumference of a circle is 22cm – Find the radius of a circle $(\pi = \frac{22}{7})$.
- A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle.
- Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm]
- Find the perimeter of a regular hexagon with sides 5cm.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Plane Shapes: Their Types, Properties and How They Differ

EXPLANATION

A plane shape is any shape that lies flat, in two dimensions — it has length and width but no thickness. Every plane shape is closed, which means its outline runs round and meets itself with no gap.

Plane shapes come in two big families.

The first family is the polygon. A polygon is a closed shape made only of straight sides. It is named by how many sides it has:
- a triangle has 3 sides,
- a quadrilateral has 4 sides,
- a pentagon has 5 sides,
- a hexagon has 6 sides,
- an octagon has 8 sides.

The second family is the circle. A circle is a closed shape made from one continuous curved line. It has no straight sides and no corners.

Inside the polygon family, a polygon is regular when every side is the same length and every angle is the same size. If the sides are not all equal or the angles are not all equal, the polygon is irregular. So a regular pentagon has 5 equal sides and 5 equal angles, while an irregular pentagon has sides of different lengths.

EXPLANATION

Every plane shape has properties — the facts about its sides, corners (vertices) and angles that are always true for that shape. When you name a shape, these are the properties that tell it apart from every other shape.

- Triangle: 3 sides, 3 vertices, 3 angles. The three angles add up to 180°.
- Quadrilateral: 4 sides, 4 vertices, 4 angles. The four angles add up to 360°. The quadrilateral family includes:
  - Square: 4 equal sides, 4 right angles, opposite sides parallel.
  - Rectangle: opposite sides equal (two pairs of equal sides), 4 right angles.
  - Parallelogram: two pairs of parallel sides, opposite sides equal, opposite angles equal.
  - Trapezium: exactly one pair of parallel sides.
- Pentagon: 5 sides, 5 vertices, 5 angles.
- Hexagon: 6 sides, 6 vertices, 6 angles.
- Octagon: 8 sides, 8 vertices, 8 angles.
- Circle: no sides and no vertices — it is one single curved edge.

Notice the pattern in the polygon family: a polygon with n sides always has n vertices and n angles. A triangle has 3 of each, an octagon has 8 of each, and so on.

WORKED_EXAMPLE

To compare two shapes you list their similarities (ways they are alike) and their differences (ways they differ). Worked examples:

Example 1 — Compare a square and a rectangle.
Similarities: Both are quadrilaterals, so both have 4 sides and 4 right angles.
Differences: A square has 4 equal sides. A rectangle has opposite sides equal, but the two pairs can be of different lengths — so a rectangle can be long and thin, while a square cannot.

Example 2 — Compare a parallelogram and a trapezium.
Similarities: Both are quadrilaterals, so both have 4 sides and 4 vertices.
Differences: A parallelogram has two pairs of parallel sides. A trapezium has exactly one pair of parallel sides.

Example 3 — Compare a triangle and an octagon.
Similarities: Both are polygons, so both are closed shapes made of straight sides.
Differences: A triangle has 3 sides, 3 vertices and its angles add to 180°. An octagon has 8 sides, 8 vertices and its angles add to a much larger total, 1080°.

1. A shape has 5 equal sides and 5 equal angles. What is its name, and is it regular or irregular?
answer: It is a regular pentagon, because all 5 sides are equal and all 5 angles are equal.

2. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all the non-adjacent vertices?

answer: 6 triangles.

3. Write down two similarities and two differences between a hexagon and a circle.
answer: Similarities: both are closed plane shapes. Differences: a hexagon has 6 straight sides and 6 vertices, while a circle has one continuous curved edge and no vertices.

4. State one similarity and one difference between a square and a parallelogram.
answer: Similarity: both are quadrilaterals with two pairs of parallel sides. Difference: a square has 4 equal sides and 4 right angles, whereas a parallelogram need not have equal sides or right angles (only opposite sides equal and opposite angles equal).

Parts of a Circle

EXPLANATION

A circle is a plane shape made of all the points at the same distance from a fixed point. That fixed point is the centre of the circle. To describe a circle fully, we name its different parts. Let's go through them one at a time.

Centre — the fixed point in the middle, the same distance from every point on the circle.

Radius — a straight line from the centre to any point on the circle. Every radius of one circle is the same length. Its name is often shortened to r.

Diameter — a straight line that passes through the centre and touches the circle at both ends. The diameter is twice the radius: d = 2r, so the radius is half the diameter: r = d ÷ 2.

Chord — a straight line that joins any two points on the circle. Notice that the diameter is a special chord: it is the longest chord because it passes through the centre.

Arc — part of the curve of the circle itself, a curved "piece" of the circumference between two points.

Circumference — the whole distance around the circle, the full curve. (You will use it later to find the perimeter of a circle.)

Segment — the region between a chord and the arc it cuts off. It looks like a "slice" with a straight edge (the chord) and a curved edge (the arc).

Sector — the region between two radii and the arc between them, shaped like a slice of pie, with the centre at its point.

Tangent — a straight line that touches the circle at exactly one point, never passing through the inside of the circle.

WORKED_EXAMPLE

Let's look at a circle and name each part you meet.

Imagine a circle drawn on paper with a dot in the middle. The dot is the **centre**.

Draw a straight line from the centre to the edge of the circle — that line is a **radius**. If the line starts at the centre but goes all the way across to the other edge, passing through the centre, it is a **diameter**. The diameter is exactly twice as long as the radius.

Now draw a straight line that joins two points on the edge but does not go through the centre. That line is a **chord**. It is shorter than the diameter because it misses the centre.

The curved part of the edge between two points is an **arc**. The whole curved edge all the way around is the **circumference**.

Shade the region between a chord and its arc — that flat, crescent-like part is a **segment**. Shade the region between two radii and the arc between them, like a wedge of cake with the centre at the point — that is a **sector**.

Finally, draw a straight line that just touches the circle at one single point and stops — it does not cut through the circle. That line is a **tangent**.

1. Name the part of a circle that is a straight line from the centre to any point on the circle.
answer: Radius.

2. A circle has a radius of 5 cm. How long is its diameter?
answer: 10 cm (diameter = 2 × radius = 2 × 5 cm).

3. What is the name of the whole distance around a circle?
answer: The circumference.

4. A straight line that touches a circle at exactly one point and does not pass through the circle is called a _____.
answer: Tangent.

5. What is the difference between a chord and a diameter?
answer: Both join two points on the circle, but a diameter passes through the centre and is the longest chord; a chord does not pass through the centre and is shorter.

6. A chord cuts off a region between itself and the arc of the circle. What is that region called?
answer: A segment.

Objective 3

Finding the Perimeter of Plane Shapes

EXPLANATION

The **perimeter** of a plane shape is the total distance around its outside. For any polygon — a shape with straight sides — you find the perimeter by adding up the lengths of all its sides.

Some shapes have equal sides, which lets you multiply instead of adding everything. A **regular polygon** has all its sides equal and all its angles equal, so its perimeter is:

perimeter of a regular polygon = number of sides × length of one side

A **square** is a regular polygon with 4 equal sides, so:

perimeter of a square = 4 × side

A **rectangle** has opposite sides equal — two lengths and two breadths (widths) — so:

perimeter of a rectangle = 2 × length + 2 × breadth

A **parallelogram** also has opposite sides equal (a base and a slanted side), and a **trapezium** has four sides that are usually all different, so for both you simply add the four side lengths.

A **circle** has no straight sides. Its perimeter is called its **circumference**:

circumference = π × diameter   or   circumference = 2 × π × radius

Remember that the diameter is twice the radius. For this work we use π = 22/7.

WORKED_EXAMPLE

Worked example 1 — adding or multiplying the sides.

A square has sides of 8 cm. Perimeter = 4 × 8 = 32 cm.

A rectangle has length 7 cm and breadth 4 cm. Perimeter = 2 × 7 + 2 × 4 = 14 + 8 = 22 cm.

A parallelogram has a base of 6 cm and a side of 5 cm. Its opposite sides are equal, so the sides are 6, 5, 6, 5. Perimeter = 6 + 5 + 6 + 5 = 22 cm.

A trapezium has sides 3 cm, 5 cm, 4 cm and 6 cm. Perimeter = 3 + 5 + 4 + 6 = 18 cm.

A regular hexagon has sides of 5 cm. It has 6 equal sides, so Perimeter = 6 × 5 = 30 cm.

WORKED_EXAMPLE

Worked example 2 — the circle and finding a missing measurement.

A circle has a radius of 7 cm. Its circumference = 2 × (22/7) × 7 = 2 × 22 = 44 cm.

Now suppose you are given the perimeter and must find a missing side or the radius. Work backwards.

A rectangle has a perimeter of 24 cm and a breadth of 4 cm. We have 2 × length + 2 × 4 = 24, so 2 × length + 8 = 24, giving 2 × length = 16, so the length is 8 cm.

The circumference of a circle is 22 cm. Use circumference = 2 × π × radius:
22 = 2 × (22/7) × r, so 22 = (44/7) × r. Multiply both sides by 7/44: r = 22 × (7/44) = 7/2 = 3.5 cm.

1. Find the perimeter of a square with sides of 8 cm.
answer: 32 cm

2. Find the perimeter of a rectangle of length 7 cm and breadth 4 cm.
answer: 22 cm

3. What is the perimeter of a regular pentagon with sides of 5 cm?

answer: 25 cm

4. Find the perimeter of a parallelogram whose base is 6 cm and sides are 5 cm.

answer: 22 cm

5. The circumference of a circle is 22 cm. Find its radius. Use $\pi = 22/7$.

answer: 3.5 cm

6. A rectangle is such that its length is twice its breadth. If its breadth is d, find the total length of the sides of the rectangle.

answer: 6d

7. Calculate the perimeter of the figure below, which is a rectangle (9 cm long, 5 cm high) with a semicircle (radius 3.5 cm) set into its top edge. The two straight top edges are each 1 cm long, and the semicircle sits between them.

answer: 32 cm

8. Find the perimeter of a regular hexagon with sides of 5 cm.

answer: 30 cm

C2 `C1 plus the lesson note as source evidence.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Plane Shapes (2-D Shapes)

`subtopic`

Plane shapes: types, properties and perimeter

`term`

Third Term

`week`

2

`lesson_objectives`

• identify the types of plane shapes and state their properties, similarities and differences
• identify the parts of a circle
• find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`entry_behaviour`

• Students have been taught angles.

`instructional_materials`

• Cardboard containing angles presentation.

Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

objective_text

identify the types of plane shapes and state their properties, similarities and differences

current_step

1

instructional_events

• present_content
• elicit_performance
• present_content

assessment_points

• How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Hints\n\nPlane shapes are also known as two dimensional shapes (2-D shapes).
They include: triangle,\nrectangle, squares, parallelogram, trapezium, circle and other polygons.
\n\n#### Triangles\n\nIsosceles: a single tick on each of the two slanted sides, an arc at each
base angle. Scalene:\none, two and three ticks on the three sides, an arc at each vertex.
Equilateral: a single tick\non each of the three sides, an arc at each vertex.\n\n[drawing
attached to this section here, unlabelled, marked Isosceles, Scalene, Equilateral]\n\n| Isosceles
| Scalene | Equilateral |\n|---|---|---|\n| (i) 2 adjacent side are equal | No side equal | 3
sides are equal |\n| (ii) base angles are equal | No angle is equal to the other | All angles are
equal ($60^\\circ$ each) |\n| (iii) one line of symmetry | No line of symmetry. | 3 lines of
symmetry. |\n\n#### Squares and Rectangles\n\nSquare: a right-angle mark at each corner, a single
tick on each of the four sides. Rectangle: a\nright-angle mark at each corner, a single tick on
each of the two longer sides and a double tick\non each of the two shorter sides.\n\n[drawing
attached to this section here, unlabelled, marked Squares, Rectangles]\n\n| Squares | Rectangles
|\n|---|---|\n| i) All sides are equal | 2 opposite sides are equal |\n| (ii) 2 pairs of parallel
sides | 2 pair of parallel sides. |\n| (iii) All angles are equal ($90^\\circ$) each. | All
angles are equal ($90^\\circ$) each. |\n| (iv) Four lines of symmetry | Two lines of symmetry |
\n| (v) Diagonals bisect it at right angle. | Diagonals bisect it but not at right angle. |
\n\n#### Parallelograms and Trapezium\n\nParallelogram: a double arrow on each of the two
horizontal sides, a single arrow on each of the\ntwo slanted sides. Trapezium: a single arrow on
each of the two parallel sides, and one diagonal\ndrawn in.\n\n[drawing attached to this section
here, unlabelled, marked Parallelogram, Trapezium]\n\n| Parallelogram | Trapezium |\n|---|---|\n|
i) 2 pairs of opposite side are equal and parallel | Only a pair of sides are parallel |\n| (ii)
opposite angles are equal | opposite angles are not equal |\n| (iii) No line of symmetry | No
line of symmetry |\n| (iv) Diagonals divide it into equal triangles | Diagonal divide it into two
unequal triangles. |\n\n#### Polygons\n\nA polygon is a closed shape that has three or more
straight sides.\n\n| Name of Polygon | No of sides | No of triangle ($n-2$) |\n|---|---|---|\n|
Triangle | 3 | 1 |\n| Quadrilateral | 4 | 2 |\n| Pentagon | 5 | 3 |\n| Hexagon | 6 | 4 |\n|
Heptagon | 7 | 5 |\n| Octagon | 8 | 6 |\n| Nonagon | 9 | 7 |\n| Decagon | 10 | 8 |\n| Hendecagon
or undecagon | 11 | 9 |\n| Dodecagon | 12 | 10 |"
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]
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Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, source_figures, objective_learner_state`.

Objective 2 — identify the parts of a circle

`objective_text`

identify the parts of a circle

`current_step`

2

`instructional_events`

- present_content

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Parts of a circle\n\nTwo labelled circles. The first has the diameter drawn
across it and a radius from its centre to\nthe circumference, with the arc and the circumference
labelled. The second has a chord near the\ntop with the segment above it shaded, a shaded sector
below with its right-hand bounding line\nlabelled Radius, and a tangent touching it on the right.
\n\n[drawing attached to this section here, unlabelled, marked Diameter, Radius, arc,
Circumference, Chord, Segment, Sector, Tangent]\n\n- **Arc:** This is a curved part of the
circumference.\n- **Radius:** This is any straight line joining the centre of the circle to the
circumference.\n- **Chord:** This is any straight line joining two point on the circumference of
a circle.\n- **Diameter:** Any chord that passes through the centre of the circle."
 }
]
```


Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

objective_text

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

current_step

3

assessment_points

• Find the perimeter of a square of sides 8cm
• Find the perimeter of a rectangle of length 7cm and breadth 4cm.
• What is the perimeter of regular pentagon with sides 5cm.
• Find the perimeter of a parallelogram whose base is 6cm and sides 5cm
• The circumference of a circle is 22cm – Find the radius of a circle $(\pi = \frac{22}{7})$.
• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle.
• Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm]
• Find the perimeter of a regular hexagon with sides 5cm.

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Perimeter\n\nPerimeter is the distance round a shape.\n\n**Worked
examples**\n\n##### 1. Calculate the perimeter of a square of side 15cm.\n\n[drawing attached to
this section here, unlabelled, marked Square, L 15cm, 15cm, L]\n\n**Solution**\n\nPerimeter of a
square $= L + L + L + L = 4L$\n\n$= 15 + 15 + 15 + 15 = 4(15)$\n\n$= 60\\mathrm{cm}$\n\n##### 2.
Find the perimeter of a rectangle whose length 10cm and breadth 7cm.\n\n[drawing attached to this
section here, unlabelled, marked L 10cm, L, b, 7cm]\n\n**Solution**\n\nPerimeter of rectangle $=
L + b + L + b$\n\n$= L + L + b + b$\n\n$= 2L + 2b$\n\n$= 2(L + B)$\n\n$= 10\\mathrm{cm} + 7\
\mathrm{cm} + 10\\mathrm{cm} + 7\\mathrm{cm}$\n\n$= 10\\mathrm{cm} + 10\\mathrm{cm} + 7\
\mathrm{cm} + 7\\mathrm{cm}$\n\n$= 2(10) + 2(7)$\n\n$= 2(10 + 7)$\n\n$= 2(17)$\n\n$= 34\
\mathrm{cm}$\n\n##### 3. If the perimeter of a rectangle is 60cm and its length is 17cm.
Calculate its breadth.\n\n[drawing attached to this section here, unlabelled, marked L = 17 cm,
b]\n\n**Solution**\n\nPerimeter $= 60\\mathrm{cm}$\n\n$P = L + b + L + b$\n\n$60 = 17\\mathrm{cm}
+ b + 17\\mathrm{cm} + b$\n\n$60 = 17\\mathrm{cm} + 17\\mathrm{cm} + b + b$\n\n$60\\mathrm{cm} =
2(17\\mathrm{cm}) + 2b$\n\n$60\\mathrm{cm} = 2(17\\mathrm{cm} + b)$\n\n$60\\mathrm{cm} = 34\
\mathrm{cm} + 2b$\n\ncollect like terms.\n\n$60\\mathrm{cm} - 34\\mathrm{cm} = 2b$\n\n$2b = 26\
\mathrm{cm}$\n\nDivide both side by 2\n\n$\\frac{2b}{2} = \\frac{26\\mathrm{cm}}{2}$\n\n$b = 13\
\mathrm{cm}$\n\n##### 4. Find the perimeter of the trapezium below.\n\n[drawing attached to this
section here, unlabelled, marked a 11cm, c 5cm, 9cm, b, 4cm, d]\n\nPerimeter of trapezium $= a +
b + c + d$\n\n$= 11\\mathrm{cm} + 9\\mathrm{cm} + 5\\mathrm{cm} + 4\\mathrm{cm}$\n\n$= 29\
\mathrm{cm}$\n\n##### 5. Find the perimeter of the parallelogram below.\n\n[drawing attached to
this section here, unlabelled, marked 9cm a, a, b, 5cm]\n\n**Solution**\n\nPerimeter of
parallelogram $= a + a + b + b$\n\n$= 2a + 2b$\n\n$= 2(a + b)$\n\n$= 2(9 + 5)$\n\n$= 2(14)$\n\n$=
28\\mathrm{cm}$\n\n##### 6. Find the perimeter of a regular hexagon sides 5.2cm.
\n\n**Solution.**\n\n[drawing attached to this section here, unlabelled, marked Regular hexagon,
5.2cm]\n\nHexagon has 6 sides.\n\nPerimeter $= 6 \\times 5.2\\mathrm{cm}$\n\n$= 31.2\\mathrm{cm}$
\n\n##### 7. Calculate the perimeter of circle of radius 7cm (use the value $3\\frac{1}{7}$ for $
\\pi$)\n\n**Solution**\n\n[drawing attached to this section here, unlabelled, marked 7cm]
\n\nPerimeter of a circle $= 2\\pi r$ or $\\pi d$\n\n$= 2 \\times 3\\frac{1}{7} \\times 7\
\mathrm{cm}$\n\n$= 2 \\times \\frac{22}{7} \\times 7$\n\n$= 2 \\times 22 = 44\\mathrm{cm}$
\n\nusing $\\pi d$.\n\nSince $r = 7\\mathrm{cm}$\n\n$d = 2 \\times r = 2 \\times 7\\mathrm{cm} =
14\\mathrm{cm}$\n\n$\\frac{22}{7} \\times 14\\mathrm{cm}$\n\n$= 22 \\times 2$\n\n$= 44\
\mathrm{cm}$\n\n##### 8. A rope is wound 50 times round a cylinder of radius 25cm. How long is
```

```
the rope? $\\pi = 3.14$.\n\n**Solution**\n\n[drawing attached to this section here, unlabelled]
\n\nlength of the rope $=$ circumference.\n\nCircumference $= 2\\pi r$\n\n$= 2 \\times 3.14 \
\times 25$\n\n$= 157\\mathrm{cm}$\n\nTotal length of the rope $= 157\\mathrm{cm} \\times 50$
\n\n$= 7850\\mathrm{cm}$.\n\n##### 9. The circumference of a circle is 440cm. Find the radius $(\
\pi = \\frac{22}{7})$\n\n**Solution**\n\n[drawing attached to this section here, unlabelled,
marked r]\n\nCircumference $= 2\\pi r$\n\n$440 = 2 \\times \\frac{22}{7} \\times r$\n\n$440 = \
\frac{44r}{7}$\n\ncross multiply.\n\n$440 \\times 7 = 44r$\n\n$3080 = 44r$\n\ndivide both side by
$44$\n\n$\\frac{44r}{44} = \\frac{3080}{44}$\n\n$r = 70\\mathrm{cm}$\n\n##### 10. Calculate the
perimeter of the shape below.\n\n[drawing attached to this section here, unlabelled, marked 4.9
cm, 4.9cm]\n\n**Solution**\n\nlength of the curved part $= \\frac{1}{4} \\times 2\\pi r$\n\n$= \
\frac{1}{4} \\times 2 \\times \\frac{22}{7} \\times 4.9$\n\n$= \\frac{1}{4} \\times \\frac{44}{7}
\\times 4.9$\n\n$= \\frac{11}{7} \\times \\frac{49}{10}$\n\n$= \\frac{11 \\times 7}{10}$\n\n$= \
\frac{77}{10}$\n\n$= 7.7\\mathrm{cm}$\n\nPerimeter of the shape $= 4.9 + 4.9 + 7.7$\n\n$= 17.5\
\mathrm{cm}$"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Plane Shapes: Types, Properties and How They Compare

EXPLANATION

A plane shape is a flat, two-dimensional (2-D) shape — it has length and width but no thickness. Plane shapes you meet all the time include the triangle, the rectangle, the square, the parallelogram, the trapezium, the circle and other polygons.

This section helps you name the different types of plane shapes, say what is special about each one, and compare them — spotting what is the same and what is different.

EXPLANATION

A triangle is a closed shape with three straight sides. Triangles come in three types, and you can tell them apart by their sides and angles (see fig-01):

Isosceles triangle — two adjacent (neighbouring) sides are equal. Its base angles are equal, and it has one line of symmetry.

Scalene triangle — no two sides are equal. No angle is equal to any other, so it has no line of symmetry.

Equilateral triangle — all three sides are equal. All three angles are equal (60° each), and it has three lines of symmetry.

So the sides tell you the type: equal sides in pairs means isosceles, no equal sides means scalene, and all sides equal means equilateral.

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(4,0)--(2,4)--cycle;
  \draw (0.8,1.6)--(1.2,1.4);
  \draw (3.2,1.6)--(2.8,1.4);
  \draw (0,0)++(0:0.7) arc[start angle=0, end angle=63, radius=0.7];
  \draw (4,0)++(180:0.7) arc[start angle=180, end angle=117, radius=0.7];
  \node[below] at (2,-0.6) {Isosceles};
  \begin{scope}[xshift=7cm]
    \draw (0,0)--(5,0)--(1.4,3.4)--cycle;
    \draw (0.38,1.83)--(1.02,1.57);
    \draw (3.02,1.87)--(3.50,2.38);
    \draw (3.38,1.53)--(3.86,2.04);
    \foreach \d in {-0.25,0,0.25} { \draw (2.5+\d,-0.25)--(2.5+\d,0.25); }
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=68, radius=0.7];
    \draw (5,0)++(180:0.8) arc[start angle=180, end angle=137, radius=0.8];
    \draw (1.4,3.4)++(-113:0.7) arc[start angle=-113, end angle=-25, radius=0.7];
    \node[below] at (2.5,-0.6) {Scalene};
  \end{scope}
  \begin{scope}[xshift=14cm]
    \draw (0,0)--(4,0)--(2,3.46)--cycle;
    \draw (0.8,1.55)--(1.2,1.32);
    \draw (3.2,1.55)--(2.8,1.32);
    \draw (1.8,-0.25)--(2.2,0.25);
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=60, radius=0.7];
    \draw (4,0)++(180:0.7) arc[start angle=180, end angle=120, radius=0.7];
    \draw (2,3.46)++(-120:0.7) arc[start angle=-120, end angle=-60, radius=0.7];
    \node[below] at (2,-0.6) {Equilateral};
  \end{scope}
\end{tikzpicture}
```

EXPLANATION

Squares and rectangles look alike, so it helps to line up their properties side by side (see fig-02):

Square — all four sides are equal; two pairs of parallel sides; all angles are equal (90° each); four lines of symmetry; its diagonals bisect it at a right angle.

Rectangle — two opposite sides are equal (the two longer sides equal each other, and the two shorter sides equal each other); two pairs of parallel sides; all angles are equal (90° each); two lines of symmetry; its diagonals bisect it but not at a right angle.

The similarity: both have two pairs of parallel sides and four right angles. The difference: the square has all sides equal, while the rectangle only has opposite sides equal; the square has four lines of symmetry, the rectangle has two; and the square's diagonals cut at a right angle while the rectangle's do not.

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (3.4,3.4);
  \draw (0,0.5)--(0.5,0.5)--(0.5,0);
  \draw (3.4,0.5)--(2.9,0.5)--(2.9,0);
  \draw (0,2.9)--(0.5,2.9)--(0.5,3.4);
  \draw (3.4,2.9)--(2.9,2.9)--(2.9,3.4);
  \draw (1.7,-0.25)--(1.7,0.25);
  \draw (1.7,3.15)--(1.7,3.65);
  \draw (-0.25,1.7)--(0.25,1.7);
  \draw (3.15,1.7)--(3.65,1.7);
  \node[below] at (1.7,-0.9) {Squares};
  \begin{scope}[xshift=8cm]
    \draw (0,0) rectangle (5,3);
    \draw (0,0.5)--(0.5,0.5)--(0.5,0);
    \draw (5,0.5)--(4.5,0.5)--(4.5,0);
    \draw (0,2.5)--(0.5,2.5)--(0.5,3);
    \draw (5,2.5)--(4.5,2.5)--(4.5,3);
    \draw (2.5,-0.25)--(2.5,0.25);
    \draw (2.5,2.75)--(2.5,3.25);
    \draw (-0.25,1.35)--(0.25,1.35); \draw (-0.25,1.65)--(0.25,1.65);
    \draw (4.75,1.35)--(5.25,1.35); \draw (4.75,1.65)--(5.25,1.65);
    \node[below] at (2.5,-0.9) {Rectangles};
  \end{scope}
\end{tikzpicture}
```

EXPLANATION

A parallelogram and a trapezium are both four-sided shapes, but they differ in how their sides sit (see fig-03):

Parallelogram — two pairs of opposite sides are equal and parallel; opposite angles are equal; no line of symmetry; its diagonals divide it into two equal triangles.

Trapezium — only one pair of sides is parallel; opposite angles are not equal; no line of symmetry; its diagonal divides it into two unequal triangles.

The key difference is the sides: the parallelogram has two pairs of parallel sides, while the trapezium has only one pair. That single difference drives everything else — the trapezium's opposite angles are unequal and its diagonal makes unequal triangles.

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(5,0)--(6.6,3)--(1.6,3)--cycle;
  \draw[->] (2.4,0)--(2.6,0);
  \draw[->] (3.9,3)--(4.1,3);
  \draw[->] (5.7,1.4)--(5.85,1.65);
  \draw[->] (5.85,1.4)--(6.0,1.65);
  \draw[->] (0.7,1.4)--(0.85,1.65);
  \draw[->] (0.85,1.4)--(1.0,1.65);
  \node[below] at (3,-0.6) {Parallelogram};
  \begin{scope}[xshift=10cm]
    \draw (0,0)--(6,0)--(4.6,3)--(1.6,3)--cycle;
    \draw (0,0)--(4.6,3);
    \draw[->] (2.9,0)--(3.1,0);
```

```
    \draw[->] (3.0,3)--(3.2,3);
    \node[below] at (3,-0.6) {Trapezium};
  \end{scope}
\end{tikzpicture}
```

EXPLANATION

A polygon is a closed shape with three or more straight sides. Each polygon has a name that tells you how many sides it has, and you can work out how many triangles it splits into by subtracting 2 from the number of sides (sides – 2):

Polygon	Number of sides	Number of triangles (n – 2)
Triangle	3	1
Quadrilateral	4	2
Pentagon	5	3
Hexagon	6	4
Heptagon	7	5
Octagon	8	6
Nonagon	9	7
Decagon	10	8
Hendecagon (undecagon)	11	9
Dodecagon	12	10

So the square, rectangle, parallelogram and trapezium you met above are all quadrilaterals — four-sided polygons.

WORKED_EXAMPLE

Let's compare two shapes you have just read about — a square and a parallelogram. Both are quadrilaterals (four sides), and both have two pairs of parallel sides, so they have a lot in common. Here is the full comparison:

Square: all four sides equal, all angles 90°, four lines of symmetry.
Parallelogram: opposite sides equal and parallel, opposite angles equal, no line of symmetry.

Same: both have two pairs of parallel sides, and both have their opposite sides equal.
Different: the square has all four sides equal and four right angles and four lines of symmetry, while the parallelogram only needs opposite sides equal and opposite angles equal and has no line of symmetry.

So the square is really a special parallelogram — a parallelogram with all sides equal and all angles right angles.

1. A triangle has all three of its sides equal and each angle measures 60°. What type of triangle is it, and how many lines of symmetry does it have?

answer: It is an equilateral triangle, and it has three lines of symmetry.

2. A four-sided shape has two pairs of parallel sides and all four sides equal, but its angles are not right angles. What is this shape called?

answer: It is a parallelogram (a parallelogram whose four sides are equal).

3. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all the non-adjacent vertices? (Hint: subtract 2 from the number of sides.)

answer: 6 triangles.

4. Name one similarity and one difference between a rectangle and a trapezium.

answer: Similarity: both are four-sided (quadrilaterals). Difference: the rectangle has two pairs of parallel sides and four right angles, while the trapezium has only one pair of parallel sides and no right angles in general.

The Parts of a Circle

EXPLANATION

Every circle is made up of a few parts that have their own names. When you look at a circle, these are the parts you can point to. Look at the two labelled circles in fig-04 and match each part to its name as you read on.

Circumference. This is the whole curved line that forms the round edge of the circle. It is the distance all the way around the circle.

Arc. An arc is a curved part of the circumference — just a piece of that round edge, not the whole thing. If you cut the circle at two points on its edge, the curved stretch of edge between them is an arc.

Centre. The point exactly in the middle of the circle, from which every point on the circumference is the same distance away. It is not labelled on the figure, but everything else is measured from it.

Radius. Any straight line that joins the centre of the circle to the circumference. In the first circle of fig-04, the line drawn straight down from the centre to the edge is a radius.

Chord. Any straight line that joins two points on the circumference of the circle. It does not need to pass through the centre — in the second circle, the straight line across the top is a chord.

Diameter. A chord that passes through the centre of the circle. In the first circle, the line drawn straight across the middle through the centre is the diameter. A diameter is always the longest chord of the circle.

Segment. The region of the circle cut off by a chord. In the second circle, the shaded part above the chord is the segment — it is the piece of the circle between the chord and the arc above it.

Sector. A part of the circle shaped like a slice, bounded by two radii and the arc between them. In the second circle, the shaded slice is the sector, and you can see its two straight edges are radii meeting at the centre.

Tangent. A straight line that touches the circle at exactly one point and does not cut across it. In the second circle, the straight vertical line on the right touches the edge of the circle at a single point — that is a tangent.

A useful fact to remember: the diameter is twice the radius. If you know the radius, the diameter is two of them; if you know the diameter, the radius is half of it.

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (-3,0)--(3,0);
  \node[above] at (0,0.1) {Diameter};
  \draw (0,0)--(0,-3);
  \node[right] at (0.45,-1.9) {Radius};
  \draw[->] (-5.9,-3.4)--(-2.3,-1.9);
  \node[left] at (-5.9,-3.4) {arc};
  \draw[->] (4.4,3.4)--(2.2,2.2);
  \node[right] at (4.4,3.4) {Circumference};
  \begin{scope}[xshift=17cm]
    \draw (0,0) circle (3);
    \draw (-2.29,1.94)--(2.29,1.94);
    \begin{scope}
      \clip (0,0) circle (3);
      \clip (-3,1.94) rectangle (3,3);
      \foreach \x in {-3,-2.7,...,3} { \draw (\x,1.8)--(\x+1.2,3.0); }
    \end{scope}
    \node[below] at (0,1.9) {Chord};
    \draw[->] (3.2,4.2)--(0.6,2.5);
    \node[right] at (3.2,4.2) {Segment};
    \begin{scope}
```

```
    \clip (0,0)--(-80:3) arc[start angle=-80, end angle=-25, radius=3]--cycle;
    \foreach \y in {-3.4,-3.1,...,0} { \draw (0,\y)--(3,\y+1.2); }
  \end{scope}
  \draw (0,0)--(-80:3);
  \draw (0,0)--(-25:3) node[midway, sloped, above] {Radius};
  \draw[->] (-3.6,-4.2)--(1.0,-2.0);
  \node[left] at (-3.6,-4.2) {Sector};
  \draw (3,-2.4)--(3,2.4);
  \node[right] at (3.2,0) {Tangent};
 \end{scope}
\end{tikzpicture}
```

WORKED_EXAMPLE

Let us work through a circle together and name its parts.

Suppose a circle has its centre marked O, and a straight line is drawn from O to a point A on the edge.

- The line OA joins the centre to the circumference, so it is a **radius**.
- If we extend OA all the way across the circle to a point B on the opposite edge, the full line AB passes through the centre, so AB is a **diameter**. It is made of two radii — OA and OB — and it is twice the length of one radius.
- Now take two points P and Q on the edge that are not opposite each other, and join them with a straight line. Since PQ joins two points on the circumference but does not pass through the centre, it is a **chord**.
- The curved stretch of the edge from P to Q is an **arc**.
- The flat region between the chord PQ and the arc above it is a **segment**.
- If the centre O is joined to P and to Q with two straight lines, the slice between the two radii OP and OQ and the arc between them is a **sector**.

So the same circle has all these parts at once: a circumference forming the edge, arcs along that edge, a radius and a diameter through the centre, a chord, a segment, a sector, and a tangent where a straight line just touches the edge at one point.

1. A straight line is drawn from the centre of a circle to a point on its circumference. What is this line called?

answer: Radius.

2. A straight line joins two points on the circumference of a circle but does not pass through the centre. Is it a chord, a diameter, or an arc?

answer: A chord.

3. A straight line touches a circle at exactly one point without cutting across it. What is this line called?

answer: A tangent.

4. A circle has a radius of 4 cm. What is its diameter? What is the radius of a circle whose diameter is 14 cm?

answer: A radius of 4 cm gives a diameter of 8 cm. A diameter of 14 cm gives a radius of 7 cm.

5. Name the three parts of a circle shown here: the whole curved edge of the circle; a curved part of that edge; and the region of the circle cut off by a chord.

answer: The whole curved edge is the circumference; a curved part of it is an arc; the region cut off by a chord is a segment.

Finding the Perimeter of Plane Shapes

EXPLANATION
The **perimeter** of a shape is the distance right round it — the total length of its boundary. To find a perimeter, you add the lengths of all the sides that go around the shape. Let's meet the perimeter of each plane shape you need.

EXPLANATION
Square. All four sides of a square are equal. If one side is L, then the perimeter is $L + L + L + L = 4L$. So the perimeter of a square is just four times one side.

Rectangle. A rectangle has two equal long sides (length L) and two equal short sides (breadth b). Its perimeter is $L + b + L + b = 2L + 2b = 2(L + b)$.

Parallelogram. A parallelogram also has two pairs of equal opposite sides: the two bases a and the two slanted sides b. Its perimeter is $a + a + b + b = 2a + 2b = 2(a + b)$.

Trapezium. A trapezium's four sides are usually all different, so its perimeter is just the sum of all four sides: $a + b + c + d$.

Regular polygon. A *regular* polygon has all sides equal. So a regular hexagon (6 sides), with each side s, has perimeter $6s$. In general a regular polygon with n equal sides has perimeter $n \times s$.

EXPLANATION
Circle. The perimeter of a circle has its own name: the **circumference**. You find it with $2\pi r$ (where r is the radius) or πd (where d is the diameter, and $d = 2r$). When a circle or part of a circle is joined to straight sides, you add the straight sides to the curved part to get the whole perimeter.

WORKED_EXAMPLE
Example 1 — Perimeter of a square
A square has side 15cm. Its perimeter is $4L = 4(15) = 60\text{cm}$.

WORKED_EXAMPLE
Example 2 — Perimeter of a rectangle
A rectangle has length 10cm and breadth 7cm. Its perimeter is $2(L + b) = 2(10 + 7) = 2(17) = 34\text{cm}$.

Example 3 — Finding a missing breadth
The perimeter of a rectangle is 60cm and its length is 17cm. Find the breadth.
\[\nP = 2L + 2b\n\]
\[\n60 = 2(17) + 2b\n\]
\[\n60 = 34 + 2b\n\]
Collect like terms:
\[\n2b = 60 - 34 = 26\n\]
Divide both sides by 2:
\[\nb = \frac{26}{2} = 13\text{cm}\n\]

WORKED_EXAMPLE
Example 4 — Perimeter of a trapezium
A trapezium has sides $a = 11$cm, $b = 9$cm, $c = 5$cm and $d = 4$cm. Its perimeter is the sum of all four sides:
\[\na + b + c + d = 11 + 9 + 5 + 4 = 29\text{cm}\n\]

Example 5 — Perimeter of a parallelogram
A parallelogram has base $a = 9$cm and slanted side $b = 5$cm. Its perimeter is:
\[\n2(a + b) = 2(9 + 5) = 2(14) = 28\text{cm}\n\]

WORKED_EXAMPLE
Example 6 — Perimeter of a regular hexagon
A regular hexagon has each side 5.2cm and has 6 sides, so:
\[\nP = 6 \times 5.2 = 31.2\text{cm}\n\]

Example 7 — Circumference of a circle
A circle has radius 7cm. Using $\pi = \tfrac{22}{7}$:
\[\nP = 2\pi r = 2 \times \frac{22}{7} \times 7 = 2 \times 22 = 44\text{cm}\n\]
(You get the same answer with πd: since $d = 2 \times 7 = 14$cm, $\frac{22}{7} \times 14 = 44$cm.)

WORKED_EXAMPLE

Example 8 — Finding a missing radius
The circumference of a circle is 440cm. Find its radius, with $\pi = \tfrac{22}{7}$.
\[\nC = 2\pi r\n\]
\[\n440 = 2 \times \frac{22}{7} \times r = \frac{44r}{7}\n\]
Cross multiply:
\[\n440 \times 7 = 44r\n\]
\[\n3080 = 44r\n\]
Divide both sides by 44:
\[\nr = \frac{3080}{44} = 70\text{cm}\n\]

WORKED_EXAMPLE
Example 9 — Perimeter of a shape with a curved part
A quarter circle has two straight sides each 4.9cm and a curved arc. With $\pi = \tfrac{22}{7}$, the curved part is one quarter of a full circumference:
\[\n\frac{1}{4} \times 2\pi r = \frac{1}{4} \times 2 \times \frac{22}{7} \times 4.9 = \frac{1}{4} \times \frac{44}{7} \times 4.9 = 7.7\text{cm}\n\]
The perimeter is the two straight sides plus the curved part:
\[\n4.9 + 4.9 + 7.7 = 17.5\text{cm}\n\]

VISUAL
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (5,5);
  \node[left]  at (-1.4,4.4) {Square};
  \node[above] at (2.5,5) {$L$\quad 15cm};
  \node[below] at (2.5,0) {$L$\quad 15cm};
  \node[left]  at (0,3) {15cm};
  \node[left]  at (0,1.8) {$L$};
  \node[right] at (5,3) {15cm};
  \node[right] at (5,1.8) {$L$};
\end{tikzpicture}
```

VISUAL
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4.5);
  \node[above] at (3.5,4.5) {$L$\quad 10cm};
  \node[below] at (3.5,0) {$L$};
  \node[left]  at (0,2.25) {$b$};
  \node[right] at (7,2.7) {$b$};
  \node[right] at (7,1.6) {7cm};
\end{tikzpicture}
```

VISUAL
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4);
  \node[above] at (3.5,4) {$L = 17$cm};
  \node[below] at (3.5,0) {$L = 17$cm};
  \node[left]  at (0,2) {$b$};
  \node[right] at (7,2) {$b$};
\end{tikzpicture}
```

VISUAL
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(7.5,0)--(6.0,3.2)--(2.0,3.2)--cycle;
  \node[below] at (3.75,0) {$a$\quad 11cm};
  \node[above] at (4.0,3.2) {$c$\quad 5cm};
  \node[left]  at (0.95,2.3) {9cm};
  \node[left]  at (0.65,1.35) {$b$};
  \node[right] at (6.9,2.3) {4cm};
  \node[right] at (7.15,1.35) {$d$};
\end{tikzpicture}
```

VISUAL
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(6,0)--(8,3.2)--(2,3.2)--cycle;
```

```
  \draw[->] (2.7,0)--(2.9,0);
  \draw[->] (4.7,3.2)--(4.9,3.2);
  \draw[->] (0.9,1.5)--(1.05,1.75);
  \draw[->] (6.9,1.5)--(7.05,1.75);
  \node[below] at (3,0) {9cm\quad $a$};
  \node[above] at (5,3.2) {$a$};
  \node[left]  at (0.9,1.6) {$b$};
  \node[right] at (7.2,1.9) {$b$};
  \node[right] at (7.4,1.1) {5cm};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \foreach \i in {0,...,5} { \coordinate (H\i) at ({\i*60}:3); }
  \draw (H0)--(H1)--(H2)--(H3)--(H4)--(H5)--cycle;
  \node[above]       at (0,4.0) {Regular hexagon};
  \node[above]       at (0,2.65) {5.2cm};
  \node[above right] at (2.4,1.3) {5.2cm};
  \node[below right] at (2.4,-1.3) {5.2cm};
  \node[below]       at (0,-2.65) {5.2cm};
  \node[below left]  at (-2.4,-1.3) {5.2cm};
  \node[above left]  at (-2.4,1.3) {5.2cm};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (0,0)--(3,0) node[midway,above] {7cm};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) ellipse (2 and 0.6);
  \draw (-2,0)--(-2,-6);
  \draw (2,0)--(2,-6);
  \draw (-2,-6) arc[start angle=180, end angle=360, x radius=2, y radius=0.6];
  \foreach \y in {-0.9,-1.8,-2.7,-3.6,-4.5,-5.4} {
    \draw (-2,\y) arc[start angle=180, end angle=360, x radius=2, y radius=0.55];
    \draw (2,\y-0.03)--(-2,\y-0.9);
  }
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (0,0)--(3,0) node[midway,above] {$r$};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(0,5)--(0,5) arc[start angle=90, end angle=0, radius=5]--(0,0);
  \node[left]  at (0,2.5) {4.9 cm};
  \node[below] at (2.5,0) {4.9cm};
\end{tikzpicture}
```

1. Find the perimeter of a square with sides 8cm.
answer: 32cm

2. Find the perimeter of a rectangle of length 7cm and breadth 4cm.
answer: 22cm

3. What is the perimeter of a regular pentagon with sides 5cm?
answer: 25cm

4. Find the perimeter of a parallelogram whose base is 6cm and slanted sides are 5cm.
answer: 22cm

5. The circumference of a circle is 22cm. Find its radius, using $\pi = \frac{22}{7}$.
answer: 3.5cm

6. A rectangle is such that its length is twice its breadth. If the breadth is d, find the total length of the sides of the rectangle.
answer: $6d$

7. Find the perimeter of a regular hexagon with sides 5cm.
answer: 30cm

C3 `C2 plus a validation and repair pass over the draft.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Plane Shapes (2-D Shapes)

`subtopic`

Plane shapes: types, properties and perimeter

`term`

Third Term

`week`

2

`lesson_objectives`

• identify the types of plane shapes and state their properties, similarities and differences
• identify the parts of a circle
• find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`entry_behaviour`

• Students have been taught angles.

`instructional_materials`

• Cardboard containing angles presentation.

Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

objective_text

identify the types of plane shapes and state their properties, similarities and differences

current_step

1

instructional_events

- present_content
- elicit_performance
- present_content

assessment_points

- How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Hints\n\nPlane shapes are also known as two dimensional shapes (2-D shapes).
They include: triangle,\nrectangle, squares, parallelogram, trapezium, circle and other polygons.
\n\n#### Triangles\n\nIsosceles: a single tick on each of the two slanted sides, an arc at each
base angle. Scalene:\none, two and three ticks on the three sides, an arc at each vertex.
Equilateral: a single tick\non each of the three sides, an arc at each vertex.\n\n[drawing
attached to this section here, unlabelled, marked Isosceles, Scalene, Equilateral]\n\n| Isosceles
| Scalene | Equilateral |\n|---|---|---|\n| (i) 2 adjacent side are equal | No side equal | 3
sides are equal |\n| (ii) base angles are equal | No angle is equal to the other | All angles are
equal ($60^\\circ$ each) |\n| (iii) one line of symmetry | No line of symmetry. | 3 lines of
symmetry. |\n\n#### Squares and Rectangles\n\nSquare: a right-angle mark at each corner, a single
tick on each of the four sides. Rectangle: a\nright-angle mark at each corner, a single tick on
each of the two longer sides and a double tick\non each of the two shorter sides.\n\n[drawing
attached to this section here, unlabelled, marked Squares, Rectangles]\n\n| Squares | Rectangles
|\n|---|---|\n| i) All sides are equal | 2 opposite sides are equal |\n| (ii) 2 pairs of parallel
sides | 2 pair of parallel sides. |\n| (iii) All angles are equal ($90^\\circ$) each. | All
angles are equal ($90^\\circ$) each. |\n| (iv) Four lines of symmetry | Two lines of symmetry |
\n| (v) Diagonals bisect it at right angle. | Diagonals bisect it but not at right angle. |
\n\n#### Parallelograms and Trapezium\n\nParallelogram: a double arrow on each of the two
horizontal sides, a single arrow on each of the\ntwo slanted sides. Trapezium: a single arrow on
each of the two parallel sides, and one diagonal\ndrawn in.\n\n[drawing attached to this section
here, unlabelled, marked Parallelogram, Trapezium]\n\n| Parallelogram | Trapezium |\n|---|---|\n|
i) 2 pairs of opposite side are equal and parallel | Only a pair of sides are parallel |\n| (ii)
opposite angles are equal | opposite angles are not equal |\n| (iii) No line of symmetry | No
line of symmetry |\n| (iv) Diagonals divide it into equal triangles | Diagonal divide it into two
unequal triangles. |\n\n#### Polygons\n\nA polygon is a closed shape that has three or more
straight sides.\n\n| Name of Polygon | No of sides | No of triangle ($n-2$) |\n|---|---|---|\n|
Triangle | 3 | 1 |\n| Quadrilateral | 4 | 2 |\n| Pentagon | 5 | 3 |\n| Hexagon | 6 | 4 |\n|
Heptagon | 7 | 5 |\n| Octagon | 8 | 6 |\n| Nonagon | 9 | 7 |\n| Decagon | 10 | 8 |\n| Hendecagon
or undecagon | 11 | 9 |\n| Dodecagon | 12 | 10 |"
 }
]
```

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, source_figures, objective_learner_state`.

Objective 2 — identify the parts of a circle

`objective_text`

identify the parts of a circle

`current_step`

2

`instructional_events`

- present_content

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Parts of a circle\n\nTwo labelled circles. The first has the diameter drawn
across it and a radius from its centre to\nthe circumference, with the arc and the circumference
labelled. The second has a chord near the\ntop with the segment above it shaded, a shaded sector
below with its right-hand bounding line\nlabelled Radius, and a tangent touching it on the right.
\n\n[drawing attached to this section here, unlabelled, marked Diameter, Radius, arc,
Circumference, Chord, Segment, Sector, Tangent]\n\n- **Arc:** This is a curved part of the
circumference.\n- **Radius:** This is any straight line joining the centre of the circle to the
circumference.\n- **Chord:** This is any straight line joining two point on the circumference of
a circle.\n- **Diameter:** Any chord that passes through the centre of the circle."
 }
]
```


Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

objective_text

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

current_step

3

assessment_points

• Find the perimeter of a square of sides 8cm
• Find the perimeter of a rectangle of length 7cm and breadth 4cm.
• What is the perimeter of regular pentagon with sides 5cm.
• Find the perimeter of a parallelogram whose base is 6cm and sides 5cm
• The circumference of a circle is 22cm – Find the radius of a circle $(\pi = \frac{22}{7})$.
• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle.
• Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm]
• Find the perimeter of a regular hexagon with sides 5cm.

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Perimeter\n\nPerimeter is the distance round a shape.\n\n**Worked
examples**\n\n##### 1. Calculate the perimeter of a square of side 15cm.\n\n[drawing attached to
this section here, unlabelled, marked Square, L 15cm, 15cm, L]\n\n**Solution**\n\nPerimeter of a
square $= L + L + L + L = 4L$\n\n$= 15 + 15 + 15 + 15 = 4(15)$\n\n$= 60\\mathrm{cm}$\n\n##### 2.
Find the perimeter of a rectangle whose length 10cm and breadth 7cm.\n\n[drawing attached to this
section here, unlabelled, marked L 10cm, L, b, 7cm]\n\n**Solution**\n\nPerimeter of rectangle $=
L + b + L + b$\n\n$= L + L + b + b$\n\n$= 2L + 2b$\n\n$= 2(L + B)$\n\n$= 10\\mathrm{cm} + 7\
\mathrm{cm} + 10\\mathrm{cm} + 7\\mathrm{cm}$\n\n$= 10\\mathrm{cm} + 10\\mathrm{cm} + 7\
\mathrm{cm} + 7\\mathrm{cm}$\n\n$= 2(10) + 2(7)$\n\n$= 2(10 + 7)$\n\n$= 2(17)$\n\n$= 34\
\mathrm{cm}$\n\n##### 3. If the perimeter of a rectangle is 60cm and its length is 17cm.
Calculate its breadth.\n\n[drawing attached to this section here, unlabelled, marked L = 17 cm,
b]\n\n**Solution**\n\nPerimeter $= 60\\mathrm{cm}$\n\n$P = L + b + L + b$\n\n$60 = 17\\mathrm{cm}
+ b + 17\\mathrm{cm} + b$\n\n$60 = 17\\mathrm{cm} + 17\\mathrm{cm} + b + b$\n\n$60\\mathrm{cm} =
2(17\\mathrm{cm}) + 2b$\n\n$60\\mathrm{cm} = 2(17\\mathrm{cm} + b)$\n\n$60\\mathrm{cm} = 34\
\mathrm{cm} + 2b$\n\ncollect like terms.\n\n$60\\mathrm{cm} - 34\\mathrm{cm} = 2b$\n\n$2b = 26\
\mathrm{cm}$\n\nDivide both side by 2\n\n$\\frac{2b}{2} = \\frac{26\\mathrm{cm}}{2}$\n\n$b = 13\
\mathrm{cm}$\n\n##### 4. Find the perimeter of the trapezium below.\n\n[drawing attached to this
section here, unlabelled, marked a 11cm, c 5cm, 9cm, b, 4cm, d]\n\nPerimeter of trapezium $= a +
b + c + d$\n\n$= 11\\mathrm{cm} + 9\\mathrm{cm} + 5\\mathrm{cm} + 4\\mathrm{cm}$\n\n$= 29\
\mathrm{cm}$\n\n##### 5. Find the perimeter of the parallelogram below.\n\n[drawing attached to
this section here, unlabelled, marked 9cm a, a, b, 5cm]\n\n**Solution**\n\nPerimeter of
parallelogram $= a + a + b + b$\n\n$= 2a + 2b$\n\n$= 2(a + b)$\n\n$= 2(9 + 5)$\n\n$= 2(14)$\n\n$=
28\\mathrm{cm}$\n\n##### 6. Find the perimeter of a regular hexagon sides 5.2cm.
\n\n**Solution.**\n\n[drawing attached to this section here, unlabelled, marked Regular hexagon,
5.2cm]\n\nHexagon has 6 sides.\n\nPerimeter $= 6 \\times 5.2\\mathrm{cm}$\n\n$= 31.2\\mathrm{cm}$
\n\n##### 7. Calculate the perimeter of circle of radius 7cm (use the value $3\\frac{1}{7}$ for $
\\pi$)\n\n**Solution**\n\n[drawing attached to this section here, unlabelled, marked 7cm]
\n\nPerimeter of a circle $= 2\\pi r$ or $\\pi d$\n\n$= 2 \\times 3\\frac{1}{7} \\times 7\
\mathrm{cm}$\n\n$= 2 \\times \\frac{22}{7} \\times 7$\n\n$= 2 \\times 22 = 44\\mathrm{cm}$
\n\nusing $\\pi d$.\n\nSince $r = 7\\mathrm{cm}$\n\n$d = 2 \\times r = 2 \\times 7\\mathrm{cm} =
14\\mathrm{cm}$\n\n$\\frac{22}{7} \\times 14\\mathrm{cm}$\n\n$= 22 \\times 2$\n\n$= 44\
\mathrm{cm}$\n\n##### 8. A rope is wound 50 times round a cylinder of radius 25cm. How long is
```

```
the rope? $\\pi = 3.14$.\n\n**Solution**\n\n[drawing attached to this section here, unlabelled]
\n\nlength of the rope $=$ circumference.\n\nCircumference $= 2\\pi r$\n\n$= 2 \\times 3.14 \
\times 25$\n\n$= 157\\mathrm{cm}$\n\nTotal length of the rope $= 157\\mathrm{cm} \\times 50$
\n\n$= 7850\\mathrm{cm}$.\n\n##### 9. The circumference of a circle is 440cm. Find the radius $(\
\pi = \\frac{22}{7})$\n\n**Solution**\n\n[drawing attached to this section here, unlabelled,
marked r]\n\nCircumference $= 2\\pi r$\n\n$440 = 2 \\times \\frac{22}{7} \\times r$\n\n$440 = \
\frac{44r}{7}$\n\ncross multiply.\n\n$440 \\times 7 = 44r$\n\n$3080 = 44r$\n\ndivide both side by
$44$\n\n$\\frac{44r}{44} = \\frac{3080}{44}$\n\n$r = 70\\mathrm{cm}$\n\n##### 10. Calculate the
perimeter of the shape below.\n\n[drawing attached to this section here, unlabelled, marked 4.9
cm, 4.9cm]\n\n**Solution**\n\nlength of the curved part $= \\frac{1}{4} \\times 2\\pi r$\n\n$= \
\frac{1}{4} \\times 2 \\times \\frac{22}{7} \\times 4.9$\n\n$= \\frac{1}{4} \\times \\frac{44}{7}
\\times 4.9$\n\n$= \\frac{11}{7} \\times \\frac{49}{10}$\n\n$= \\frac{11 \\times 7}{10}$\n\n$= \
\frac{77}{10}$\n\n$= 7.7\\mathrm{cm}$\n\nPerimeter of the shape $= 4.9 + 4.9 + 7.7$\n\n$= 17.5\
\mathrm{cm}$"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Plane Shapes: Types, Properties and How They Differ

EXPLANATION

A plane shape (also called a two-dimensional shape, or 2-D shape) is a flat shape that you can draw on a surface. The common plane shapes are the triangle, rectangle, square, parallelogram, trapezium, circle and the other polygons. In this section you will meet the most important ones, learn their properties, and see how they are similar to and different from one another.

Whenever a shape is drawn, small marks tell you about its sides and angles:

A tick (a short line across a side) marks sides that are equal. Two sides with one tick each are equal to each other; sides with two ticks are equal to each other, and so on.
A right-angle mark (a small square at a corner) shows an angle of 90°.
An arc (a curved mark at a corner) marks an angle in a diagram.
A line of symmetry is a line you could fold the shape along so that the two halves match exactly.

EXPLANATION

Triangles come in three special types, which differ in how many of their sides and angles are equal (see fig-01).

Isosceles triangle — two of its adjacent sides are equal (marked with a single tick on each slanted side). Its two base angles are equal, and it has one line of symmetry.

Scalene triangle — no two sides are equal (its three sides carry one, two and three ticks). No angle is equal to any other, and it has no line of symmetry.

Equilateral triangle — all three sides are equal (a single tick on each side). All three angles are equal, each 60°, and it has three lines of symmetry.

So the isosceles and equilateral triangles are alike in having at least two equal sides and a line of symmetry, while the scalene triangle differs from both because it has no equal sides and no line of symmetry.

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(4,0)--(2,4)--cycle;
  \draw (0.8,1.6)--(1.2,1.4);
  \draw (3.2,1.6)--(2.8,1.4);
  \draw (0,0)++(0:0.7) arc[start angle=0, end angle=63, radius=0.7];
  \draw (4,0)++(180:0.7) arc[start angle=180, end angle=117, radius=0.7];
  \node[below] at (2,-0.6) {Isosceles};
  \begin{scope}[xshift=7cm]
    \draw (0,0)--(5,0)--(1.4,3.4)--cycle;
    \draw (0.38,1.83)--(1.02,1.57);
    \draw (3.02,1.87)--(3.50,2.38);
    \draw (3.38,1.53)--(3.86,2.04);
    \foreach \d in {-0.25,0,0.25} { \draw (2.5+\d,-0.25)--(2.5+\d,0.25); }
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=68, radius=0.7];
    \draw (5,0)++(180:0.8) arc[start angle=180, end angle=137, radius=0.8];
    \draw (1.4,3.4)++(-113:0.7) arc[start angle=-113, end angle=-25, radius=0.7];
    \node[below] at (2.5,-0.6) {Scalene};
  \end{scope}
  \begin{scope}[xshift=14cm]
    \draw (0,0)--(4,0)--(2,3.46)--cycle;
    \draw (0.8,1.55)--(1.2,1.32);
    \draw (3.2,1.55)--(2.8,1.32);
    \draw (1.8,-0.25)--(2.2,0.25);
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=60, radius=0.7];
    \draw (4,0)++(180:0.7) arc[start angle=180, end angle=120, radius=0.7];
    \draw (2,3.46)++(-120:0.7) arc[start angle=-120, end angle=-60, radius=0.7];
```

```
    \node[below] at (2,-0.6) {Equilateral};
  \end{scope}
\end{tikzpicture}
```

```
EXPLANATION
```

Squares and rectangles are both four-sided shapes whose angles are all right angles, but their sides differ (see fig-02).

Square — all four sides are equal. It has two pairs of parallel sides, all four angles equal to 90°, four lines of symmetry, and its diagonals cut it at right angles.

Rectangle — only the two opposite sides are equal (the two longer sides match, and the two shorter sides match). It has two pairs of parallel sides, all four angles equal to 90°, two lines of symmetry, and its diagonals cut it in half but not at a right angle.

The square and the rectangle are alike because both have two pairs of parallel sides and four right angles. They differ because the square's four sides are all equal, while the rectangle's opposite sides are equal; and the square has four lines of symmetry while the rectangle has two.

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (3.4,3.4);
  \draw (0,0.5)--(0.5,0.5)--(0.5,0);
  \draw (3.4,0.5)--(2.9,0.5)--(2.9,0);
  \draw (0,2.9)--(0.5,2.9)--(0.5,3.4);
  \draw (3.4,2.9)--(2.9,2.9)--(2.9,3.4);
  \draw (1.7,-0.25)--(1.7,0.25);
  \draw (1.7,3.15)--(1.7,3.65);
  \draw (-0.25,1.7)--(0.25,1.7);
  \draw (3.15,1.7)--(3.65,1.7);
  \node[below] at (1.7,-0.9) {Squares};
  \begin{scope}[xshift=8cm]
    \draw (0,0) rectangle (5,3);
    \draw (0,0.5)--(0.5,0.5)--(0.5,0);
    \draw (5,0.5)--(4.5,0.5)--(4.5,0);
    \draw (0,2.5)--(0.5,2.5)--(0.5,3);
    \draw (5,2.5)--(4.5,2.5)--(4.5,3);
    \draw (2.5,-0.25)--(2.5,0.25);
    \draw (2.5,2.75)--(2.5,3.25);
    \draw (-0.25,1.35)--(0.25,1.35); \draw (-0.25,1.65)--(0.25,1.65);
    \draw (4.75,1.35)--(5.25,1.35); \draw (4.75,1.65)--(5.25,1.65);
    \node[below] at (2.5,-0.9) {Rectangles};
  \end{scope}
\end{tikzpicture}
```

```
EXPLANATION
```

Parallelogram and trapezium are both four-sided shapes, and their difference is all about parallel sides (see fig-03).

Parallelogram — it has two pairs of opposite sides that are equal and parallel (the arrows on the sides show which sides are parallel). Its opposite angles are equal, it has no line of symmetry, and its diagonals divide it into two equal triangles.

Trapezium — only one pair of sides is parallel. Its opposite angles are not equal, it has no line of symmetry, and its diagonal divides it into two unequal triangles.

So both shapes have no line of symmetry, but the parallelogram has two pairs of equal and parallel sides while the trapezium has only one pair of parallel sides.

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(5,0)--(6.6,3)--(1.6,3)--cycle;
  \draw[->] (2.4,0)--(2.6,0);
  \draw[->] (3.9,3)--(4.1,3);
```

```
  \draw[->] (5.7,1.4)--(5.85,1.65);
  \draw[->] (5.85,1.4)--(6.0,1.65);
  \draw[->] (0.7,1.4)--(0.85,1.65);
  \draw[->] (0.85,1.4)--(1.0,1.65);
  \node[below] at (3,-0.6) {Parallelogram};
  \begin{scope}[xshift=10cm]
    \draw (0,0)--(6,0)--(4.6,3)--(1.6,3)--cycle;
    \draw (0,0)--(4.6,3);
    \draw[->] (2.9,0)--(3.1,0);
    \draw[->] (3.0,3)--(3.2,3);
    \node[below] at (3,-0.6) {Trapezium};
  \end{scope}
\end{tikzpicture}
```

EXPLANATION

Polygons are closed shapes that have three or more straight sides. Each polygon is named after the number of sides it has, and a polygon with n sides can be divided into n – 2 triangles by drawing diagonals from one vertex to all the non-adjacent vertices. Here are the polygons you should know:

<table>
<tr><th>Name of polygon</th><th>Number of sides</th><th>Number of triangles (n – 2)</th></tr>
<tr><td>Triangle</td><td>3</td><td>1</td></tr>
<tr><td>Quadrilateral</td><td>4</td><td>2</td></tr>
<tr><td>Pentagon</td><td>5</td><td>3</td></tr>
<tr><td>Hexagon</td><td>6</td><td>4</td></tr>
<tr><td>Heptagon</td><td>7</td><td>5</td></tr>
<tr><td>Octagon</td><td>8</td><td>6</td></tr>
<tr><td>Nonagon</td><td>9</td><td>7</td></tr>
<tr><td>Decagon</td><td>10</td><td>8</td></tr>
<tr><td>Hendecagon (or undecagon)</td><td>11</td><td>9</td></tr>
<tr><td>Dodecagon</td><td>12</td><td>10</td></tr>
</table>

Notice the pattern: for every polygon, the number of triangles is always two less than the number of sides.

WORKED_EXAMPLE

Example 1 — Comparing a square and a rectangle
State one way a square and a rectangle are the same, and one way they differ.

Answer. They are the same because both have two pairs of parallel sides and all four angles are 90°. They differ because all four sides of a square are equal, but only the opposite sides of a rectangle are equal (and the square has four lines of symmetry while the rectangle has two).

WORKED_EXAMPLE

Example 2 — Naming a polygon and its triangles
A closed shape has eight straight sides. What is its name, and into how many triangles can it be divided?

Answer. An eight-sided polygon is an octagon. Using n – 2 with n = 8, it can be divided into 8 – 2 = 6 triangles.

1. Name the three types of triangle, and for each state how many of its sides are equal and how many lines of symmetry it has.
answer: Isosceles: 2 equal sides, 1 line of symmetry. Scalene: 0 equal sides, 0 lines of symmetry. Equilateral: 3 equal sides, 3 lines of symmetry.

2. Write down one property a square and a rectangle share, and one property that is true of a square but not of a rectangle.
answer: Shared: both have two pairs of parallel sides and all four angles are 90°. True of the square only: all four sides are equal (and the square has four lines of symmetry, while the rectangle has two).

3. A parallelogram has two pairs of opposite sides equal and parallel. A trapezium has only one pair of parallel sides. Write down one property the two shapes share.
answer: Both have no line of symmetry.

4. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?
answer: 6 triangles

Parts of a Circle

EXPLANATION

A circle is a plane shape whose boundary is a perfectly round curve. Before we can work with circles, we need to name the parts that make it up. Study the two labelled circles in **fig-04** as you read each part below.

Start with the boundary itself. The whole round curved edge of the circle is called the **circumference**. An **arc** is any curved part of the circumference — if you travel only part of the way around the circle, that curved stretch is an arc.

Now look at the straight lines that can be drawn in a circle:
- A **radius** is any straight line joining the centre of the circle to the circumference. Notice that a radius always reaches from the exact middle of the circle out to the curved edge.
- A **chord** is any straight line joining two points on the circumference. It is a straight line whose two ends both lie on the circle's edge.
- A **diameter** is any chord that passes through the centre of the circle. So a diameter is a special kind of chord — it is the widest chord, because it cuts straight through the middle from one side of the circle to the other.

Finally, the circle can be divided into regions and has a line that touches it:
- The **segment** is the region between a chord and the arc it cuts off. In the second circle of **fig-04**, the part above the chord, shaded with lines, is a segment.
- The **sector** is the region between two radii and the arc between them — it looks like a slice of the circle. In **fig-04** the shaded slice at the bottom is a sector, and you can see its two straight sides are radii.
- A **tangent** is a straight line that touches the circle at exactly one point and does not pass through it. In **fig-04** the vertical line on the right touches the circle at a single point — that is a tangent.

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (-3,0)--(3,0);
  \node[above] at (0,0.1) {Diameter};
  \draw (0,0)--(0,-3);
  \node[right] at (0.45,-1.9) {Radius};
  \draw[->] (-5.9,-3.4)--(-2.3,-1.9);
  \node[left] at (-5.9,-3.4) {arc};
  \draw[->] (4.4,3.4)--(2.2,2.2);
  \node[right] at (4.4,3.4) {Circumference};
  \begin{scope}[xshift=17cm]
    \draw (0,0) circle (3);
    \draw (-2.29,1.94)--(2.29,1.94);
    \begin{scope}
      \clip (0,0) circle (3);
      \clip (-3,1.94) rectangle (3,3);
      \foreach \x in {-3,-2.7,...,3} { \draw (\x,1.8)--(\x+1.2,3.0); }
    \end{scope}
    \node[below] at (0,1.9) {Chord};
    \draw[->] (3.2,4.2)--(0.6,2.5);
    \node[right] at (3.2,4.2) {Segment};
    \begin{scope}
      \clip (0,0)--(-80:3) arc[start angle=-80, end angle=-25, radius=3]--cycle;
      \foreach \y in {-3.4,-3.1,...,0} { \draw (0,\y)--(3,\y+1.2); }
    \end{scope}
    \draw (0,0)--(-80:3);
    \draw (0,0)--(-25:3) node[midway, sloped, above] {Radius};
    \draw[->] (-3.6,-4.2)--(1.0,-2.0);
    \node[left] at (-3.6,-4.2) {Sector};
    \draw (3,-2.4)--(3,2.4);
    \node[right] at (3.2,0) {Tangent};
```

```
  \end{scope}
\end{tikzpicture}
```

WORKED_EXAMPLE

Example: naming the parts of a circle

Look at **fig-04**. Trace each part as you read.

The first circle shows a **diameter** drawn straight across the middle, through the centre, from one side of the circumference to the other. A **radius** is drawn from the centre straight down to the circumference — it is exactly half of the diameter, because the radius reaches only from the centre to the edge. The curved stretch marked by the arrow is an **arc**, a part of the circumference, and the outer arrow points to the **circumference** itself, the whole round edge.

Now the second circle. The straight line near the top, joining two points on the circumference, is a **chord**. The shaded region above that chord, between the chord and the arc, is a **segment**. Below, the shaded slice bounded by two radii and an arc is a **sector** — one of its bounding straight lines is labelled **Radius**. And the straight line on the right, which touches the circle at only one point, is a **tangent**.

So, in one look at a circle, ask yourself: is it a line reaching from the centre to the edge (radius), a line joining two points on the edge (chord), a chord through the centre (diameter), the whole curved edge (circumference), a curved part of that edge (arc), a slice between two radii (sector), the region between a chord and its arc (segment), or a line just touching at one point (tangent)?

1. What name is given to any straight line joining the centre of a circle to the circumference?
answer: Radius.

2. A straight line joining two points on the circumference of a circle is called a _____.
answer: Chord.

3. State in one sentence what a diameter of a circle is.
answer: A diameter is any chord that passes through the centre of the circle.

4. What do we call a curved part of the circumference of a circle?
answer: An arc.

5. Name the region between a chord and the arc it cuts off.
answer: A segment.

6. A straight line that touches a circle at exactly one point is called a _____.
answer: Tangent.

7. In your own words, explain the difference between a sector and a segment of a circle.
answer: A sector is the region between two radii and the arc between them (a slice of the circle), while a segment is the region between a chord and the arc it cuts off.

Finding the Perimeter of Plane Shapes

EXPLANATION

The perimeter of a shape is the distance all the way round it. To find it you add up the lengths of every side of the shape. For a circle the perimeter has its own name — the circumference — and you find it with the special formula $2\pi r$ or πd, where r is the radius, d is the diameter (twice the radius), and π is about $\frac{22}{7}$ or 3.14. In this section you'll find the perimeter of squares, rectangles, trapeziums, parallelograms, regular polygons and circles, and you'll also work backwards to find a missing side when the perimeter is already given.

WORKED_EXAMPLE

Example 1 — a square. Calculate the perimeter of a square of side 15cm.

A square has four equal sides, so its perimeter is four times one side:

$$\text{perimeter} = L + L + L + L = 4L$$

$$= 4 \times 15\text{cm} = 60\text{cm}$$

So the perimeter of the square is 60cm.

WORKED_EXAMPLE

Example 2 — a rectangle. Find the perimeter of a rectangle whose length is 10cm and breadth is 7cm.

A rectangle has two long sides (the length L) and two short sides (the breadth b), so:

$$\text{perimeter} = L + b + L + b = 2L + 2b = 2(L + b)$$

$$= 2(10\text{cm} + 7\text{cm}) = 2 \times 17\text{cm} = 34\text{cm}$$

The perimeter of the rectangle is 34cm.

WORKED_EXAMPLE

Example 3 — finding a missing side. If the perimeter of a rectangle is 60cm and its length is 17cm, calculate its breadth.

Start from the formula and substitute what you know:

$$P = L + b + L + b$$

$$60 = 17 + b + 17 + b$$

$$60 = 2(17) + 2b = 34 + 2b$$

Collect the like terms by taking 34 from both sides:

$$60 - 34 = 2b$$

$$2b = 26$$

Divide both sides by 2:

$$\frac{2b}{2} = \frac{26}{2} = 13$$

So the breadth b is 13cm.

WORKED_EXAMPLE

Example 4 — a trapezium. Find the perimeter of a trapezium whose sides are 11cm, 9cm, 5cm and 4cm.

A trapezium is a four-sided shape, so you simply add all four sides:

$$\text{perimeter} = a + b + c + d$$

$$= 11\text{cm} + 9\text{cm} + 5\text{cm} + 4\text{cm} = 29\text{cm}$$

The perimeter of the trapezium is 29cm.

WORKED_EXAMPLE

Example 5 — a parallelogram. Find the perimeter of a parallelogram whose base is 9cm and side is 5cm.

A parallelogram has two equal long sides and two equal slanted sides, so:

$$\text{perimeter} = a + a + b + b = 2a + 2b = 2(a + b)$$

$$= 2(9\text{cm} + 5\text{cm}) = 2 \times 14\text{cm} = 28\text{cm}$$

The perimeter of the parallelogram is 28cm.

WORKED_EXAMPLE

Example 6 — a regular hexagon. Find the perimeter of a regular hexagon of side 5.2cm.

A regular hexagon has six equal sides, so its perimeter is six times one side:
$$\text{perimeter} = 6 \times 5.2\text{cm} = 31.2\text{cm}$$

The perimeter of the regular hexagon is 31.2cm.

The same idea works for any regular polygon: multiply the number of sides by the length of one side.

WORKED_EXAMPLE
Example 7 — a circle. Calculate the perimeter (circumference) of a circle of radius 7cm, using $\pi = \frac{22}{7}$.

The circumference is $2\pi r$:
$$= 2 \times \frac{22}{7} \times 7 = 2 \times 22 = 44\text{cm}$$

You can check it with πd. Since $d = 2r = 2 \times 7 = 14\text{cm}$:
$$\frac{22}{7} \times 14 = 22 \times 2 = 44\text{cm}$$

The circumference of the circle is 44cm.

WORKED_EXAMPLE
Example 8 — a rope round a cylinder. A rope is wound 50 times round a cylinder of radius 25cm. How long is the rope? Use $\pi = 3.14$.

Each complete turn of the rope goes once round the circle, so one turn is the circumference:
$$= 2\pi r = 2 \times 3.14 \times 25 = 157\text{cm}$$

The rope makes 50 turns, so its total length is:
$$157\text{cm} \times 50 = 7850\text{cm}$$

The rope is 7850cm long.

WORKED_EXAMPLE
Example 9 — finding the radius. The circumference of a circle is 440cm. Find its radius, using $\pi = \frac{22}{7}$.

Use the formula and substitute:
$$C = 2\pi r$$
$$440 = 2 \times \frac{22}{7} \times r = \frac{44r}{7}$$

Cross multiply:
$$440 \times 7 = 44r$$
$$3080 = 44r$$

Divide both sides by 44:
$$\frac{44r}{44} = \frac{3080}{44} = 70$$

So the radius of the circle is 70cm.

WORKED_EXAMPLE
Example 10 — a quarter circle. Calculate the perimeter of a shape made of two straight sides each 4.9cm joined by a quarter-circle arc, using $\pi = \frac{22}{7}$.

A full circle has circumference $2\pi r$. This arc is a quarter of a circle, so the length of the curved part is:
$$\frac{1}{4} \times 2\pi r = \frac{1}{4} \times 2 \times \frac{22}{7} \times 4.9$$
$$= \frac{1}{4} \times \frac{44}{7} \times 4.9 = \frac{11}{7} \times 4.9$$
$$= \frac{11}{7} \times \frac{49}{10} = \frac{11 \times 7}{10} = \frac{77}{10} = 7.7\text{cm}$$

Now add the two straight sides:
$$\text{perimeter} = 4.9 + 4.9 + 7.7 = 17.5\text{cm}$$

The perimeter of the shape is 17.5cm.

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (5,5);
  \node[left]  at (-1.4,4.4) {Square};
  \node[above] at (2.5,5) {$L$\quad 15cm};
  \node[below] at (2.5,0) {$L$\quad 15cm};
  \node[left]  at (0,3) {15cm};
  \node[left]  at (0,1.8) {$L$};
  \node[right] at (5,3) {15cm};
  \node[right] at (5,1.8) {$L$};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4.5);
  \node[above] at (3.5,4.5) {$L$\quad 10cm};
  \node[below] at (3.5,0) {$L$};
  \node[left]  at (0,2.25) {$b$};
  \node[right] at (7,2.7) {$b$};
  \node[right] at (7,1.6) {7cm};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4);
  \node[above] at (3.5,4) {$L = 17$cm};
  \node[below] at (3.5,0) {$L = 17$cm};
  \node[left]  at (0,2) {$b$};
  \node[right] at (7,2) {$b$};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(7.5,0)--(6.0,3.2)--(2.0,3.2)--cycle;
  \node[below] at (3.75,0) {$a$\quad 11cm};
  \node[above] at (4.0,3.2) {$c$\quad 5cm};
  \node[left]  at (0.95,2.3) {9cm};
  \node[left]  at (0.65,1.35) {$b$};
  \node[right] at (6.9,2.3) {4cm};
  \node[right] at (7.15,1.35) {$d$};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(6,0)--(8,3.2)--(2,3.2)--cycle;
  \draw[->] (2.7,0)--(2.9,0);
  \draw[->] (4.7,3.2)--(4.9,3.2);
  \draw[->] (0.9,1.5)--(1.05,1.75);
  \draw[->] (6.9,1.5)--(7.05,1.75);
  \node[below] at (3,0) {9cm\quad $a$};
  \node[above] at (5,3.2) {$a$};
  \node[left]  at (0.9,1.6) {$b$};
  \node[right] at (7.2,1.9) {$b$};
  \node[right] at (7.4,1.1) {5cm};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \foreach \i in {0,...,5} { \coordinate (H\i) at ({\i*60}:3); }
  \draw (H0)--(H1)--(H2)--(H3)--(H4)--(H5)--cycle;
  \node[above]       at (0,4.0) {Regular hexagon};
  \node[above]       at (0,2.65) {5.2cm};
  \node[above right] at (2.4,1.3) {5.2cm};
  \node[below right] at (2.4,-1.3) {5.2cm};
```

```
  \node[below]       at (0,-2.65) {5.2cm};
  \node[below left]  at (-2.4,-1.3) {5.2cm};
  \node[above left]  at (-2.4,1.3) {5.2cm};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (0,0)--(3,0) node[midway,above] {7cm};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) ellipse (2 and 0.6);
  \draw (-2,0)--(-2,-6);
  \draw (2,0)--(2,-6);
  \draw (-2,-6) arc[start angle=180, end angle=360, x radius=2, y radius=0.6];
  \foreach \y in {-0.9,-1.8,-2.7,-3.6,-4.5,-5.4} {
    \draw (-2,\y) arc[start angle=180, end angle=360, x radius=2, y radius=0.55];
    \draw (2,\y-0.03)--(-2,\y-0.9);
  }
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (0,0)--(3,0) node[midway,above] {$r$};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(0,5)--(0,5) arc[start angle=90, end angle=0, radius=5]--(0,0);
  \node[left]  at (0,2.5) {4.9 cm};
  \node[below] at (2.5,0) {4.9cm};
\end{tikzpicture}
```

1. Find the perimeter of a square of side 8cm.
answer: 32cm

2. Find the perimeter of a rectangle of length 7cm and breadth 4cm.
answer: 22cm

3. What is the perimeter of a regular pentagon with sides 5cm?
answer: 25cm

4. Find the perimeter of a parallelogram whose base is 6cm and sides are 5cm.
answer: 22cm

5. The circumference of a circle is 22cm. Find the radius of the circle ($\pi = \frac{22}{7}$).
answer: 3.5cm

6. A rectangle is such that its length is twice its breadth. If the breadth is d, find the total length of the sides of the rectangle.
answer: $6d$

7. Find the perimeter of a regular hexagon with sides 5cm.
answer: 30cm

8. A shape is made of two straight sides each 4.9cm joined by a quarter-circle arc. Calculate its perimeter ($\pi = \frac{22}{7}$).
answer: 17.5cm

Result — per rubric dimension

Scores are 1–5. Every figure is the mean of 3 draws rated by 2 judge pass(es) — 6 judgments per cell.

Rubric	C0	C1	C2	C3	C1 – C0	C2 – C1	C3 – C2
Objective and curriculum alignment	5.00	5.00	5.00	5.00	+0.00	+0.00	+0.00
Taught-lesson alignment	4.00	4.33	4.00	4.00	**+0.33**	**-0.33**	+0.00
Scope and conceptual correctness	4.67	4.67	4.17	4.33	+0.00	**-0.50**	**+0.17**
Student-facing clarity and usefulness	5.00	5.00	5.00	5.00	+0.00	+0.00	+0.00

Why the judge scored it that way

Every rationale the judge wrote for this lesson, deduplicated across draws and passes. These are its words, not a summary.

Objective and curriculum alignment — C0 5.00, C1 5.00, C2 5.00, C3 5.00

- `C0` All three lesson objectives are explicitly and thoroughly addressed in sequence. Every shape type and circle part named in the objectives appears in the pack with appropriate explanation and practice.
- `C0` Every stated objective is addressed in its own dedicated section with explanation, worked examples, and practice questions. Nothing required by the objectives is omitted.
- `C0` All three stated lesson objectives are covered systematically and completely. Every shape and circle part named in the objectives appears in the pack, and the perimeter section explicitly handles every shape type the teacher specified.
- `C0` All three lesson objectives are addressed systematically and in full. The only negligible gap is the absence of 'tangent' in the circle-parts section, but every other named curriculum element is covered with worked examples and practice items.
- `C0` All three lesson objectives are fully addressed in the same order they appear in the lesson plan, with appropriate depth for JSS1. Nothing significant from the objectives is omitted.
- `C0` All three lesson objectives are addressed in dedicated sections with worked examples and practice questions. The only minor gap is that the tangent is not discussed in the circle section, but every other named part is covered thoroughly.
- `C1` The revision pack covers all three stated objectives with appropriate depth and completeness, leaving no objective unaddressed.
- `C1` All three lesson objectives are explicitly and thoroughly covered in the pack. Every shape and circle part listed in the lesson plan appears in the revision material, and the perimeter section includes the reverse-calculation skill the teacher's evaluation questions require.
- `C1` Every stated objective is fully covered in the revision pack with appropriate depth and content. Nothing from the objectives is missing or misrepresented.
- `C1` The revision pack maps directly onto each of the three stated objectives without omission, and the depth of coverage (formulas, worked examples, backwards-calculation) matches the curriculum target for JSS1 Third Term.
- `C1` All three stated lesson objectives are addressed in full and in the same order they appear in the plan. No objective is omitted or only partially treated, giving complete curriculum coverage.
- `C1` The revision pack comprehensively addresses every stated lesson objective in sequence, with no objective omitted or under-represented. Coverage matches the curriculum target for JSS1 Third Term week 2 on plane shapes.
- `C2` The revision pack covers every stated objective fully and in appropriate depth for JSS1 Third Term curriculum. Nothing required by the lesson plan is omitted.
- `C2` The pack covers every stated objective fully and in the correct depth for JSS1 Third Term. No objective is omitted or superficially treated.
- `C2` The revision pack covers every stated objective fully and at the right depth for JSS1 Third Term. Nothing from the objectives is omitted, and no irrelevant content displaces the targets.
- `C2` All three lesson objectives are covered in full and in the correct depth for JSS1. No objective is omitted or treated superficially.
- `C2` The pack maps cleanly onto every stated objective. No objective is omitted, and the depth of treatment matches what the lesson plan specifies, including reverse-calculation problems.

- `C2` The pack maps directly onto every stated objective without omission. All shapes named in the lesson plan appear with their perimeter treatment, and the reverse-calculation skill is explicitly modelled.
- `C3` All three stated lesson objectives are addressed fully and in the correct JSS1 curriculum scope. No objective is omitted or only partially treated.
- `C3` Every stated lesson objective is covered thoroughly and in appropriate depth for JSS1. No objective is omitted, and the treatment matches the curriculum level expected for Third Term, Week 2.
- `C3` All three lesson objectives are addressed explicitly and completely. The pack maps directly onto each objective in order, and the depth of treatment matches what the curriculum target for JSS1 Third Term would require.
- `C3` All three lesson objectives are addressed thoroughly and in the correct curriculum context for JSS1 Third Term. The pack maps directly onto every stated objective with explanations, examples, and practice.
- `C3` All three lesson objectives are addressed completely and in appropriate depth for JSS1. The pack maps directly onto the stated curriculum targets without omitting any required shape or concept.
- `C3` The revision pack covers every stated objective fully and in appropriate depth for JSS1 Third Term, leaving nothing from the lesson objectives unaddressed.

Taught-lesson alignment — C0 4.00, C1 4.33, C2 4.00, C3 4.00

- `C0` The pack closely follows the taught sequence and covers the same shapes and methods the teacher used, but it omits the composite-figure perimeter task and the student-participation element from the lesson, keeping it slightly short of a full match.
- `C0` The pack closely mirrors the lesson's sequence and most of its depth, but omits the algebraic/missing-side perimeter problems and the composite figure that featured in the teacher's evaluation, so it falls slightly short of full taught-lesson alignment.
- `C0` The pack reflects the lesson's sequence, emphasis and most activities well, but it does not include any practice on finding a missing side or radius from a given perimeter, which the teacher explicitly taught and assessed, so alignment is strong but not complete.
- `C0` The pack closely follows the lesson's sequence and depth. It loses one point because the teacher's board-drawing/student-identification activity is inherently interactive and cannot be replicated in a text pack, and the lesson's explicit mention of finding a missing side for polygons other than the circle is not illustrated with a worked example.
- `C0` The pack reflects the lesson's sequence and most of its expected depth very well, but the reverse-perimeter and composite-shape problem types that the teacher explicitly evaluated are not practised by students in the pack's exercises, leaving a small gap in taught-lesson alignment.
- `C0` The sequence and emphasis closely mirror the taught lesson, and most of the board examples reappear as worked examples. The absence of reverse-perimeter problems (finding radius or side from a given perimeter) is a notable omission given the teacher explicitly modelled these.
- `C1` The pack faithfully follows the lesson's sequence and emphasis and covers essentially all the teacher's evaluation questions; the only minor departure is the additional composite-shape example which is slightly beyond the explicit lesson scope but not harmful.
- `C1` The pack closely mirrors the lesson's sequence, method and expected depth, and its practice questions align tightly with the teacher's own evaluation. A small amount of additional detail (triangle classification, octagon diagonals worked example) goes slightly beyond the stated presentation activities, but does not misrepresent the lesson.
- `C1` The pack reflects the taught lesson's sequence, emphasis and depth faithfully, including the reverse-calculation skill explicitly called out in the lesson plan. The evaluation questions are well represented.
- `C1` The pack closely mirrors the lesson's sequence and covers every evaluation question, but the composite-shape worked example and the fairly extended prose explanations of similarities/differences go a little beyond what the teacher's own notes record as the lesson's activities and depth.
- `C1` The pack tracks the lesson's sequence and depth closely. A small deduction is warranted because the pack adds triangle-type content (equilateral, isosceles, scalene, right-angled) that the teacher's plan does not explicitly address in the presentation steps, slightly expanding beyond what was demonstrably taught.
- `C1` The pack mirrors the lesson's sequence, method, and depth very closely, and its practice questions align tightly with the teacher's own evaluation questions, indicating strong taught-lesson alignment throughout.
- `C2` The pack aligns well with the taught lesson's sequence and assessment emphasis. The small addition of the composite shape and the unexplained cylinder figure are minor excursions beyond the lesson as planned, preventing a full 5.
- `C2` Sequence and emphasis closely match the planned lesson, and the teacher's own evaluation questions are almost entirely reproduced. The modest extension to composite shapes is the only notable departure, keeping the score at 4 rather than 5.

- `C2` The pack's structure and emphasis track the taught lesson well, with the same sequence and the same types of problems. The few additions (polygon table, equilateral-parallelogram note) are minor elaborations rather than departures, so alignment is high but not perfect.
- `C2` The pack faithfully reproduces the lesson's sequence and depth and echoes the teacher's own evaluation questions for perimeter, though the lesson's instructional activities (students identifying properties from board drawings) are necessarily replaced by expository text in a take-home format, which is appropriate. A minor deduction is warranted because one extra worked example (quarter-circle composite shape) goes slightly beyond what the lesson explicitly planned, but it does not misrepresent the lesson.
- `C2` Sequence, emphasis, and depth are very well aligned with the taught lesson. The quarter-circle composite shape and the cylinder illustration extend modestly beyond what was explicitly taught, which is the only reason this falls short of a 5.
- `C2` The pack reflects the taught sequence and the teacher's assessed depth closely. A minor gap is that the lesson plan mentions a composite/unlabelled figure question (question 8, dimensions 1cm/5cm/9cm) which does not appear in the pack, and the quarter-circle example added goes slightly beyond what was explicitly taught.
- `C3` The sequence and emphasis strongly reflect the lesson as planned, and the evaluation questions are reproduced almost verbatim. The few additions (quarter circle, extended polygon table) go modestly beyond what the teacher presented but do not distort the core taught content.
- `C3` The sequence and depth closely reflect the taught lesson, and the evaluation questions are well-matched. The extra worked examples (quarter circle, cylinder) slightly exceed the lesson's demonstrated scope but do not detract from alignment; the pack earns a 4 rather than 5 because these additions shift emphasis beyond what was explicitly taught.
- `C3` The pack faithfully reflects the lesson's sequence, methods and emphasis. The only minor gap is that the composite figure question (Q8 in the teacher's evaluation, referencing measurements 1cm, 5cm, 9cm on an unlabelled figure) is replaced by the quarter-circle shape, which is a reasonable substitute but not identical to what the teacher set.
- `C3` The pack closely mirrors the lesson's sequence, methods, and depth across all three sections, including reverse-calculation problems the teacher modelled. The additions (quarter-circle, extended polygon table) are minor extras that do not distort alignment but prevent a perfect score.
- `C3` The pack mirrors the teacher's sequence and emphasis very closely, and the practice questions align tightly with the teacher's evaluation questions. The small additions (quarter circle, rope problem) are consistent with the lesson's spirit but go slightly beyond what was explicitly taught, preventing a full 5.
- `C3` The sequence, method, and depth closely match the taught lesson; the minor addition of the rope and quarter-circle examples represent a small but noticeable extension beyond what the teacher explicitly covered, preventing a full 5.

Scope and conceptual correctness — C0 4.67, C1 4.67, C2 4.17, C3 4.33

- `C0` No mathematical or conceptual errors were found anywhere in the pack; all definitions, formulas, worked examples and answer keys are correct and stay within JSS1 scope.
- `C0` No mathematical or conceptual errors were found anywhere in the pack; definitions, formulas, worked examples, and answer key entries are all correct and appropriately scoped for JSS1.
- `C0` The mathematics is correct throughout and stays well within JSS1 scope; the only minor issues are the π-value inconsistency with the teacher's preferred form and the borderline classification of rectangle as irregular, neither of which is a serious conceptual error.
- `C0` Every mathematical statement, formula, and definition in the pack is conceptually correct and age-appropriate. The content stays within JSS1 scope without introducing material beyond the lesson boundary.
- `C0` No mathematical or conceptual errors were found anywhere in the pack. The scope stays firmly within what is appropriate for JSS1 third term, without introducing unnecessary advanced content.
- `C0` Mathematical content is accurate throughout, with correct formulas, worked examples and answers. The classification of a rectangle as 'irregular' is the one conceptually debatable point at JSS1 level; everything else is within scope and error-free.
- `C1` No conceptual or computational errors were found; every definition, formula and worked answer is correct and within the expected JSS1 scope.
- `C1` No mathematical or conceptual errors were found anywhere in the pack. All definitions, formulas, worked examples and answers in the 'Check your work' section are correct and appropriate for JSS1 level.
- `C1` All mathematical content is conceptually correct with no errors found. The one minor concern is the composite semicircle-rectangle perimeter question, which goes slightly beyond the shapes and level specified in the lesson plan, but does not introduce wrong ideas.
- `C1` No mathematical or conceptual errors were found anywhere in the pack; formulas, worked solutions, definitions, and relational statements are all correct and within the expected JSS1 scope.

- `C1` No mathematical or conceptual errors were found anywhere in the pack. Definitions, formulas, worked examples and answers in the answer key are all correct, and the content stays within the expected JSS1 scope.
- `C1` The mathematical content is accurate throughout and stays within the lesson's boundaries. The composite figure in question 7 extends slightly beyond what the teacher's original figure description confirms, but no conceptual or computational errors are present.
- `C2` The mathematics throughout is sound and well within JSS1 scope. The unexplained cylinder/rope figure is an odd inclusion that sits outside the lesson boundary, but it does not introduce any conceptual error.
- `C2` The mathematical content is accurate throughout with no conceptual errors detected. The mild scope extension into composite/partial-circle perimeters is the only issue, but it is handled correctly and does not mislead students.
- `C2` The mathematics is accurate throughout, and the content is mostly within scope. The unlabelled cylinder figure and the non-standard 'equilateral parallelogram' label are minor blemishes that prevent a full 5.
- `C2` The content is conceptually and computationally correct throughout; the only concern is one worked example (composite shape with a quarter arc) that goes a little beyond the lesson's stated scope, though it does not introduce a wrong concept.
- `C2` Every mathematical statement, formula, worked example, and answer-key entry checked out as correct. No conceptual errors or scope violations were found.
- `C2` Mathematical content is accurate throughout. The one concern is calling a rhombus an 'equilateral parallelogram' rather than using the standard name, which is within scope but may cause terminology confusion at this level.
- `C3` The mathematics throughout is accurate and the conceptual explanations are sound. The main scope issue is the quarter-circle topic, which the teacher did not include in the lesson plan; otherwise the pack stays within bounds.
- `C3` The mathematical content is overwhelmingly correct and appropriate. The one slight terminological imprecision (isosceles sides described as 'adjacent') and the two out-of-scope examples (quarter circle, cylinder) prevent a perfect score, but there are no conceptual errors that would mislead students.
- `C3` There are no mathematical errors in the pack. The slight scope extension (cylinder rope problem and quarter-circle) adds material beyond the teacher's stated lesson, but nothing is conceptually wrong and the extensions are mathematically sound for the level.
- `C3` The mathematical content is accurate throughout, with no errors found in formulas, worked solutions, or answers. The scope expands modestly beyond the lesson (quarter circle, extended polygon table) but nothing taught is conceptually wrong.
- `C3` No mathematical or conceptual errors were found across the explanations, worked examples, or answer key. All definitions (chord, diameter, segment, sector, tangent) are precisely stated and consistent with standard JSS1 curriculum expectations.
- `C3` No mathematical or conceptual errors were found anywhere in the pack; all definitions, formulas, worked examples and answers are correct and within JSS1 scope.

Student-facing clarity and usefulness — C0 5.00, C1 5.00, C2 5.00, C3 5.00

- `C0` The pack is exceptionally well-structured for independent home use: accessible language, logical progression, multiple worked examples per topic, practice questions matched to the examples, and a detailed answer key all combine to make it highly usable for a JSS1 student working alone.
- `C0` The pack is exceptionally well structured for independent home use: accessible language, logical progression, clear worked examples, and explanatory answer keys all combine to make it highly useful for a JSS1 learner revising alone.
- `C0` The pack is exceptionally well-structured for independent home use: accessible language, concrete analogies, modelled examples, and self-check answers make it highly usable for a JSS1 learner working alone.
- `C0` The pack is exceptionally well-structured for independent home use: friendly language, logical progression, varied practice items, and detailed answer explanations ensure a JSS1 student can work through it without adult support.
- `C0` The pack is exceptionally well-structured for independent use: clear language, logical progression, self-contained worked examples, and complete answer keys with explanations mean a JSS1 student can meaningfully review the entire lesson without teacher guidance.
- `C0` The pack is exceptionally well structured for independent home revision: explanations are simple and jargon is introduced gradually, worked examples model exactly what the practice questions require, and full solutions with reasoning allow self-checking.
- `C1` The pack is clearly structured, uses accessible language throughout, and provides all the scaffolding (definitions, worked examples, practice, answers) a JSS1 student needs to revise independently and check understanding.
- `C1` The pack is well-structured, uses plain language appropriate for JSS1, and provides enough scaffolding — definitions, examples, practice and full answers — for a student to work through it independently at home.

- C1 The pack is exceptionally well-structured for independent home use: language is simple, worked solutions are detailed and explanatory rather than just numerical, and the format is consistent and learner-friendly throughout.
- C1 The pack is exceptionally well-structured for independent home use: language is simple and consistent, the worked-example format models exactly what students need to do, and the annotated answer key allows reliable self-assessment.
- C1 The pack is very well structured for unsupported home study: clear explanations, concrete analogies, matched worked examples, and a fully worked answer key all make it highly usable for a JSS1 learner working alone.
- C1 The pack is exceptionally well-structured for independent home use: language is clear and age-appropriate, worked examples model each question type before students attempt practice, and full solutions with explanations support self-correction.
- C2 The pack is exceptionally well-structured for independent revision: clear explanations, matched worked examples, and a full answer key make it genuinely useful for a JSS1 learner working alone at home.
- C2 The pack is exceptionally well structured for unsupported home use: explanations are student-friendly, examples are fully worked, and answers with working are provided. A JSS1 learner could independently review and self-assess all three lesson topics using this material.
- C2 The pack is well-structured, uses simple language appropriate for JSS1, and gives students everything they need to revise and check themselves at home. The layout and self-marking answers make it genuinely useful as a standalone revision resource.
- C2 The pack is exceptionally well-suited for independent home use: language is age-appropriate, every concept is illustrated with a step-by-step example, and complete solutions allow self-correction. Figure descriptions are included to compensate for the take-home context.
- C2 The pack is exceptionally well-structured for independent home use: definitions are accessible, examples are step-by-step, and the answer key gives full working so a JSS1 student can diagnose their own errors.
- C2 The pack is exceptionally well structured for independent home use: plain language, labelled figures, stepped working, and a complete answer key make it highly usable for a JSS1 student revising alone.
- C3 The pack is exceptionally well structured for independent home use: clear language, consistent layout, labelled figures described in text, and a detailed answer key make it highly useful for JSS1 learners revising without a teacher.
- C3 The pack is exceptionally well-organised for independent home use: clear language, logical progression, multiple worked examples before practice, and complete answer keys. A JSS1 student should be able to work through it productively on their own.
- C3 The pack is exceptionally well structured for independent home use: clear language, labelled diagrams described in words, varied worked examples, and an answer key with explanations rather than bare answers. A JSS1 student could work through it with minimal adult support.
- C3 The pack is exceptionally well-structured for independent use: clear explanations, multiple worked examples at the right level, matched practice questions, and a detailed answer key make it highly useful as take-home revision material for JSS1 learners.
- C3 The pack is exceptionally well-structured for independent home use: clear language, ample worked examples, labelled figures, and complete answer keys make it highly usable for a JSS1 student revising without a teacher.
- C3 The pack is exceptionally well-structured for independent home use: clear explanations, fully worked examples, matched practice questions, and detailed answer keys make it highly useful for a JSS1 learner revising alone.